

From exponential smoothing methods to ETS models

04

Outline

- (10) Taxonomy of **exponential smoothing** (ETS) methods
- (30) Simple exponential smoothing (SES)
- (30) Holt and damped trend
- (10) Models with seasonality
- (10) ETS methods vs models
- (25) Business implication: Quantile forecast and inventory
- Appendix
- Theory: ETS models*
- Estimation & model selection*

Introduction

- Exponential smoothing was proposed in the late 1950s and has motivated some of the most successful forecasting methods.
 - Brown (1959): **Simple exponential smoothing (SES)**
 - **Holt** (1957): Linear exponential smoothing for data with trends
 - Holt (1957) & Winters (1960): “**Holt-Winters**’ method” for seasonal data
- Forecasts produced using exponential smoothing methods are weighted averages of past observations, with the weights decaying exponentially as the observations get older.

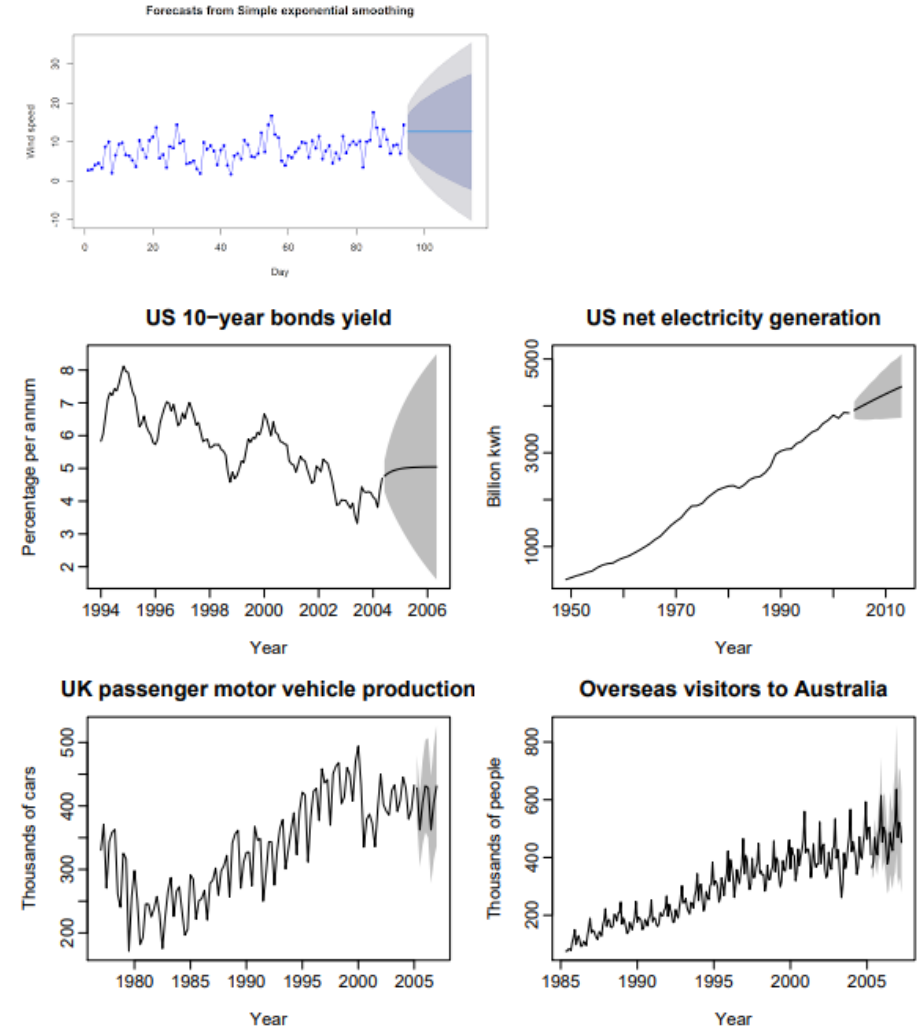


Fig. 2.1: Four time series showing point forecasts and 80% prediction intervals obtained using exponential smoothing state space models.

https://robjhyndman.com/expsmooth/expsmooth_ch2.pdf

Introduction to ETS

```
y <- tsibble(
  Quarter = yearquarter("2010 Q3") + 0:13,
  Obs = c(1212, 1321, 1278, 1341, 1257, 1341, 1257, 1287,
          1189, 1111, 1145, 1150, 1298, 1331),
  index = Quarter
)

yTrain <- y %>%
  filter(year(Quarter) <= 2012)

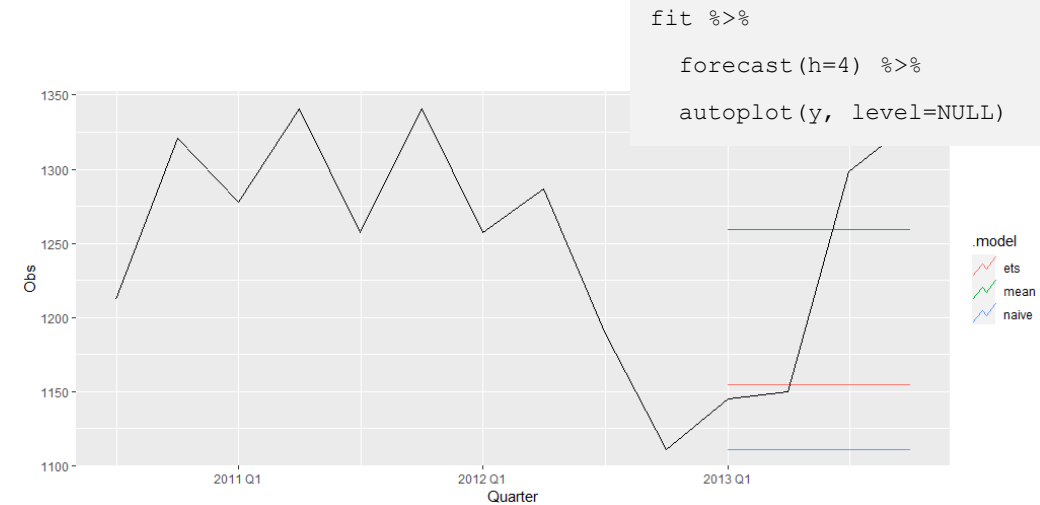
fit <- yTrain %>%
  model(ets = ETS(Obs),
        mean = MEAN(Obs),
        naive = NAIVE(Obs)
  )

fc <- fit %>%
  forecast(h=4)

accuracy(fc, y)

fit %>%
  select(ets) %>%
  report()
```

```
> fit
# A mable: 1 x 3
      ets      mean      naive
  <model> <model> <model>
1 <ETS(A,N,N)> <MEAN> <NAIVE>
```



```
> fc
# A fable: 12 x 4 [1Q]
# Key:   .model [3]
      .model Quarter      Obs .mean
   <chr>   <qtr>   <dbl> <dbl>
1 ets     2013 Q1  N(1154, 5926) 1154.
2 ets     2013 Q2  N(1154, 8205) 1154.
3 ets     2013 Q3  N(1154, 10483) 1154.
4 ets     2013 Q4  N(1154, 12762) 1154.
5 mean    2013 Q1  N(1259, 5800) 1259.
6 mean    2013 Q2  N(1259, 5800) 1259.
7 mean    2013 Q3  N(1259, 5800) 1259.
8 mean    2013 Q4  N(1259, 5800) 1259.
9 naive   2013 Q1  N(1111, 6162) 1111.
10 naive  2013 Q2  N(1111, 12323) 1111.
11 naive  2013 Q3  N(1111, 18485) 1111.
12 naive  2013 Q4  N(1111, 24647) 1111.
```

```
> accuracy(fc, y)
# A tibble: 3 x 10
      .model .type      ME      RMSE      MAE      MPE      MAPE      MASE      RMSSE      ACF1
   <chr>   <chr>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>
1 ets     Test     76.6   114.    83.5    5.78    6.38    1.14    1.11    0.290
2 mean    Test    -28.4    89.0    83.5   -2.79    6.96    1.14    0.866    0.290
3 naive   Test    120    147.    120     9.32    9.32    1.64    1.43    0.290
```

```
> fit %>%
+ select(ets) %>%
+ report()
Series: Obs Trend, Season
Model: ETS(A, N, N)
```

Smoothing parameters:
alpha = 0.6201022

Initial states:
1[0]
1249.287

sigma^2: 5925.981

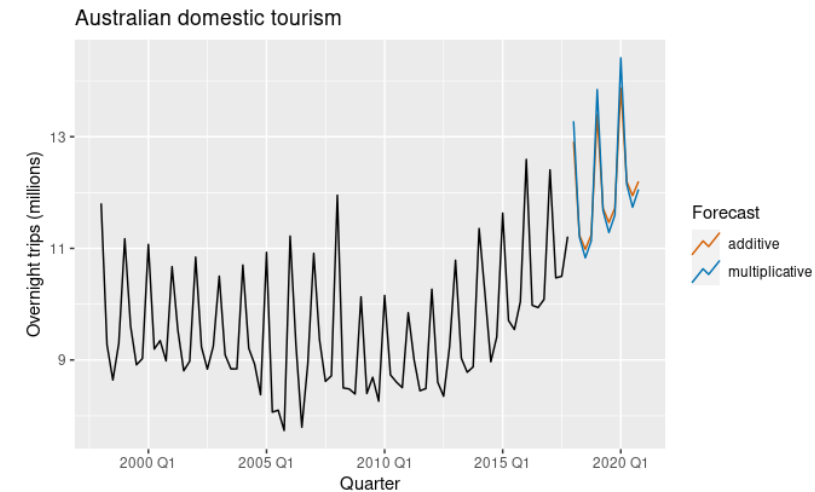
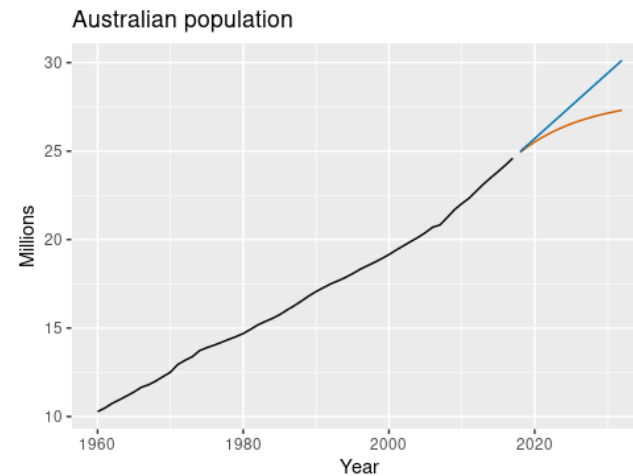
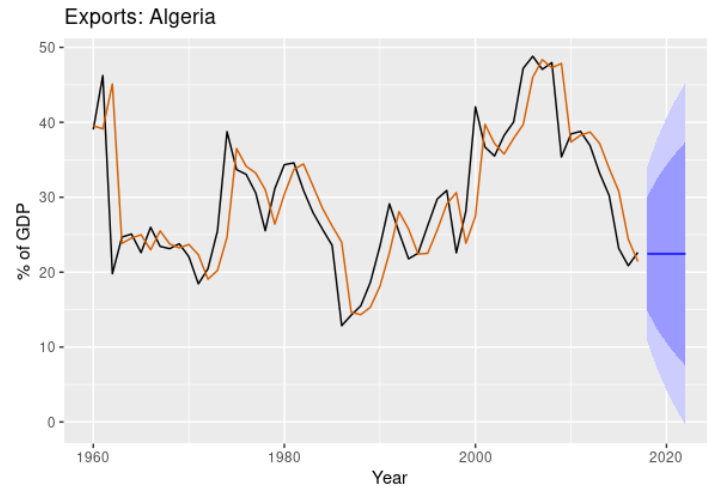
AIC AICc BIC
113.6654 117.6654 114.5732

Taxonomy of exponential smoothing methods

Trend Component	Seasonal Component		
	N	A	M
	(None)	(Additive)	(Multiplicative)
N (None)	(N,N)	(N,A)	(N,M)
A (Additive)	(A,N)	(A,A)	(A,M)
A_d (Additive damped)	(A_d ,N)	(A_d ,A)	(A_d ,M)

Short hand	Method
(N,N)	Simple exponential smoothing
(A,N)	Holt's linear method
(A_d ,N)	Additive damped trend method
(A,A)	Additive Holt-Winters' method
(A,M)	Multiplicative Holt-Winters' method
(A_d ,M)	Holt-Winters' damped method

$(T,S) = (_, _)$



Exponential smoothing method using R

```
y <- tsibble(
  Quarter = yearquarter("2010 Q3") + 0:13,
  Obs = c(1212, 1321, 1278, 1341, 1257, 1341, 1257, 1287,
          1189, 1111, 1145, 1150, 1298, 1331),
  index = Quarter
)

yTrain <- y %>%
  filter(year(Quarter) <= 2012)

yTrain %>%
  model(ets = ETS(Obs),
        mean = MEAN(Obs),
        naive = NAIVE(Obs),
        ses = ETS(Obs ~ error("A") + trend("N") + season("N")),
        holt = ETS(Obs ~ error("A") + trend("A") + season("N")),
        hw = ETS(Obs ~ error("A") + trend("A") + season("A")),
  ) %>%
  forecast(h=4) %>%
  accuracy(y)
```

Short hand	Method
(N,N)	Simple exponential smoothing
(A,N)	Holt's linear method
(A _d ,N)	Additive damped trend method
(A,A)	Additive Holt-Winters' method
(A,M)	Multiplicative Holt-Winters' method
(A _d ,M)	Holt-Winters' damped method

.model	.type	ME	RMSE	MAE	MPE	MAPE	MASE	RMSSE	ACF1
<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
ets	Test	76.6	114.	83.5	5.78	6.38	1.14	1.11	0.290
holt	Test	53.1	110.	95.3	3.80	7.47	1.30	1.07	0.300
hw	Test	60.2	133.	117.	4.27	9.25	1.61	1.30	0.140
mean	Test	-28.4	89.0	83.5	-2.79	6.96	1.14	0.866	0.290
naive	Test	120	147.	120	9.32	9.32	1.64	1.43	0.290
ses	Test	76.6	114.	83.5	5.78	6.38	1.14	1.11	0.290

FORECAST.ETS function

► *Applies To*

Important

The **FORECAST.ETS** function is not available in Excel for the web, iOS, or Android.

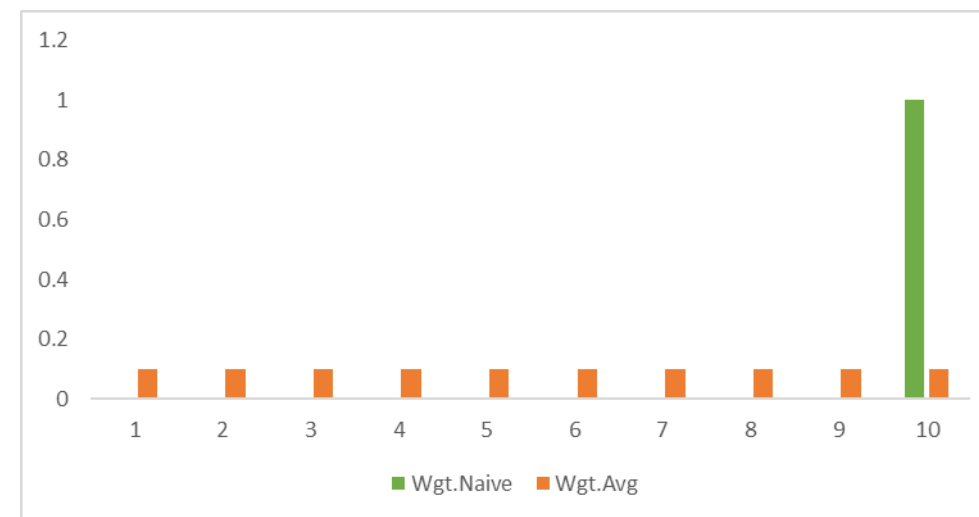
Calculates or predicts a future value based on existing (historical) values by using the AAA version of the Exponential Smoothing (ETS) algorithm. The predicted value is a continuation of the historical values in the specified target date, which should be a continuation of the timeline. You can use this function to predict future sales, inventory requirements, or consumer trends.

<https://support.microsoft.com/en-US/Excel/forecast-ets-function>

simple exponential smoothing

Review: Two extremes of weighted averages

Time period	Set	Year	Quarter	Demand	Wgt.Naive	Naïve	Wgt.Avg	Avg
$T = 10$	1	Train	2010	Q3	1212	0	0.1	
	2	Train	2010	Q4	1321	0	0.1	
	3	Train	2011	Q1	1278	0	0.1	
	4	Train	2011	Q2	1341	0	0.1	
	5	Train	2011	Q3	1257	0	0.1	
	6	Train	2011	Q4	1341	0	0.1	
	7	Train	2012	Q1	1257	0	0.1	
	8	Train	2012	Q2	1287	0	0.1	
	9	Train	2012	Q3	1189	0	0.1	
	10	Train	2012	Q4	1111	1	0.1	
	11	Test	2013	Q1	1145			1259.4
	12	Test	2013	Q2	1150			1259.4
	13	Test	2013	Q3	1298			1259.4
	14	Test	2013	Q4	1331			1259.4



- **Naïve method (random walk)**

$$\begin{aligned}\hat{y}_{T+h|T} &= y_T \\ &= 0y_1 + 0y_2 + \cdots + 0y_{T-1} + 1y_T\end{aligned}$$

- **Average (mean) method**

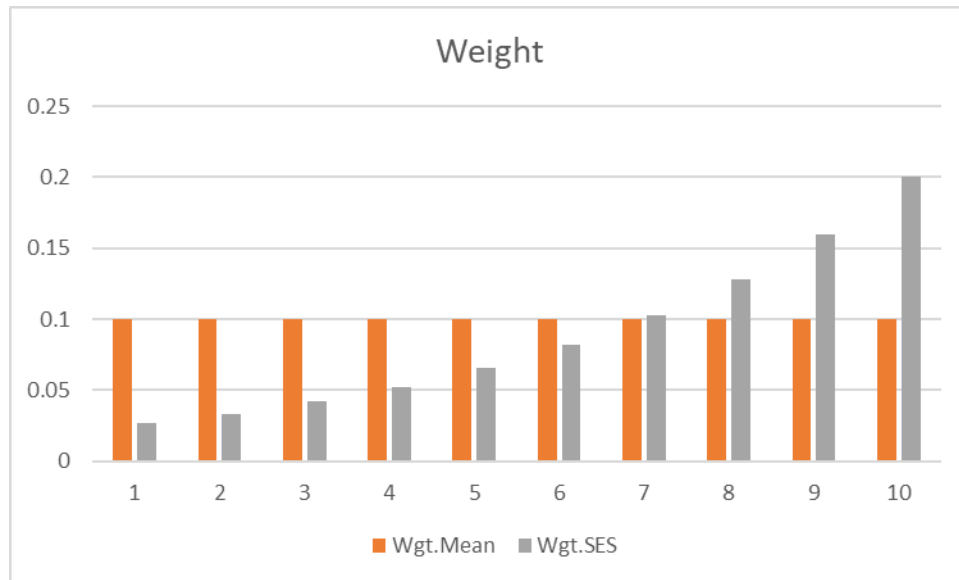
$$\begin{aligned}\hat{y}_{T+h|T} &= \frac{\sum_{i=1}^T y_i}{T} \\ &= \frac{1}{T}y_1 + \frac{1}{T}y_2 + \cdots + \frac{1}{T}y_{T-1} + \frac{1}{T}y_T\end{aligned}$$



Simple Exponential Smoothing (SES)

- Let $\alpha \in [0,1]$ be the **smoothing parameter**.
- Weights decrease **exponentially**.
 - Sum of the weights $1 - (1 - \alpha)^T \approx 1$ if T large

$$\hat{y}_{T+h|T} = \alpha y_T + \alpha(1 - \alpha)y_{T-1} + \alpha(1 - \alpha)^2 y_{T-2} + \dots$$



		$\alpha = 0.2$	
Time period	Demand	Wgt.SES	SES
1	1212	0.0268	
2	1321	0.0336	
3	1278	0.0419	
4	1341	0.0524	
5	1257	0.0655	
6	1341	0.0819	
7	1257	0.1024	
8	T-2	0.1280	
9	T-1	0.1600	
10	T=10	0.2000	
11	T+1		1098.9
12	T+2		1098.9
13			1098.9
14			1098.9

SES: Smoothing parameter $0 \leq \alpha \leq 1$

$$\hat{y}_{T+h|T} = \alpha y_T + \alpha(1 - \alpha)y_{T-1} + \alpha(1 - \alpha)^2 y_{T-2} + \dots$$

Larger α , more weight on recent observations.

	$\alpha = 0.2$	$\alpha = 0.4$	$\alpha = 0.6$	$\alpha = 0.8$
y_T	0.2000	0.4000	0.6000	0.8000
y_{T-1}	0.1600	0.2400	0.2400	0.1600
y_{T-2}	0.1280	0.1440	0.0960	0.0320
y_{T-3}	0.1024	0.0864	0.0384	0.0064
y_{T-4}	0.0819	0.0518	0.0154	0.0013
y_{T-5}	0.0655	0.0311	0.0061	0.0003

If $\alpha = 1$, $\hat{y}_{T+h|T} =$
mean or naïve methods?

SES: recursive form

$$\hat{y}_{T+h|T} = \alpha y_T + (1 - \alpha) \hat{y}_{T|T-1}$$

= weighted average b/w observation at time T & one-step forecast for time T

Recall

$$\begin{aligned}\hat{y}_{T+h|T} &= \alpha y_T + \alpha(1 - \alpha)y_{T-1} + \alpha(1 - \alpha)^2 y_{T-2} + \dots \\ &= \alpha y_T + (1 - \alpha)[\alpha y_{T-1} + \alpha(1 - \alpha)y_{T-2} + \dots] \\ &= \alpha y_T + (1 - \alpha) \hat{y}_{T|T-1}\end{aligned}$$

since

$$\hat{y}_{T|T-1} = \alpha y_{T-1} + \alpha(1 - \alpha)y_{T-2} + \alpha(1 - \alpha)^2 y_{T-3} + \dots$$

SES: component form

- Let ℓ_t be the **level** of the series at time t .
- The component form of SES is
 - **Forecast equation** $\hat{y}_{t+h|t} = \ell_t$
 - **Smoothing (level) equation** $\ell_t = \alpha y_t + (1 - \alpha)\ell_{t-1}$
- We obtain the recursive form, if we replace
 - ℓ_t with $\hat{y}_{t+1|t}$
 - ℓ_{t-1} with $\hat{y}_{t|t-1}$

Simple exponential smoothing (SES)

- Let ℓ_t be the level of the series at time t .
 - ℓ_0 is the initial level (state)
- Let $\alpha \in [0,1]$ be the **smoothing parameter**.
- The component form of SES is

- Forecast equation

$$\hat{y}_{t+h|t} = \ell_t$$

- Smoothing equation

$$\ell_t = \alpha y_t + (1 - \alpha)\ell_{t-1}$$

$\ell_0 = 1249.287$
 $\alpha = 0.6201022$

$\ell_1 =$

```
> fit %>%
+   select(ets) %>%
+   report()
Series: Obs
Model: ETS(A,N,N)
Smoothing parameters:
  alpha = 0.6201022

Initial states:
  l[0]
1249.287
```

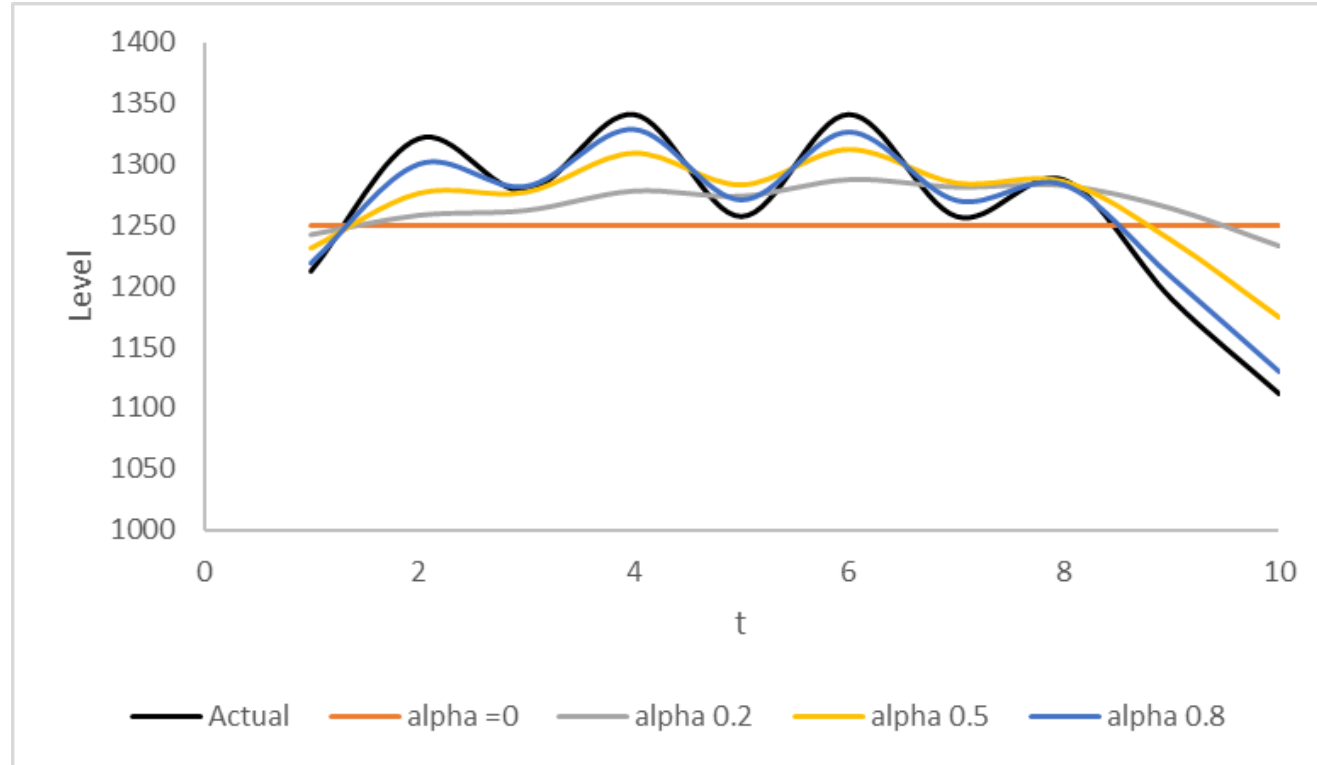
```
> fc %>%
+   filter(.model=='ets')
# A fable: 4 x 4 [1Q]
# Key:      .model [1]
      .model Quarter      Obs .mean
  <chr>    <qtr>      <dbl> <dbl>
1 ets     2013 Q1  N(1154, 5926) 1154.
2 ets     2013 Q2  N(1154, 8205) 1154.
3 ets     2013 Q3  N(1154, 10483) 1154.
4 ets     2013 Q4  N(1154, 12762) 1154.
```



Simple exponential smoothing (SES)			
Initial Level (ℓ_0)		1249.287	
Smoothing parameter (α between 0,1)		0.6201022	
Time	Actual (y_t)	Level (ℓ_t)	Forecast
0		1249.287	
1	1212	1226.165	
2	1321	1284.972	
3	1278	1280.649	
4	1341	1318.073	
5	1257	1280.201	
6	1341	1317.903	
7	1257	1280.137	
8	1287	1284.393	
9	1189	1225.239	
10	1111	1154.399	
11			1154
12			1154
13			1154
14			1154



Effect of smoothing parameter $\alpha \in [0,1]$



Initial level $\ell_0 = 1259.4$

- α controls the flexibility of level
- If $\alpha = 0$, the level never updates (fixed at initial level ℓ_0)
- If $\alpha = 1$, the level updates completely to the latest observation (naïve method)

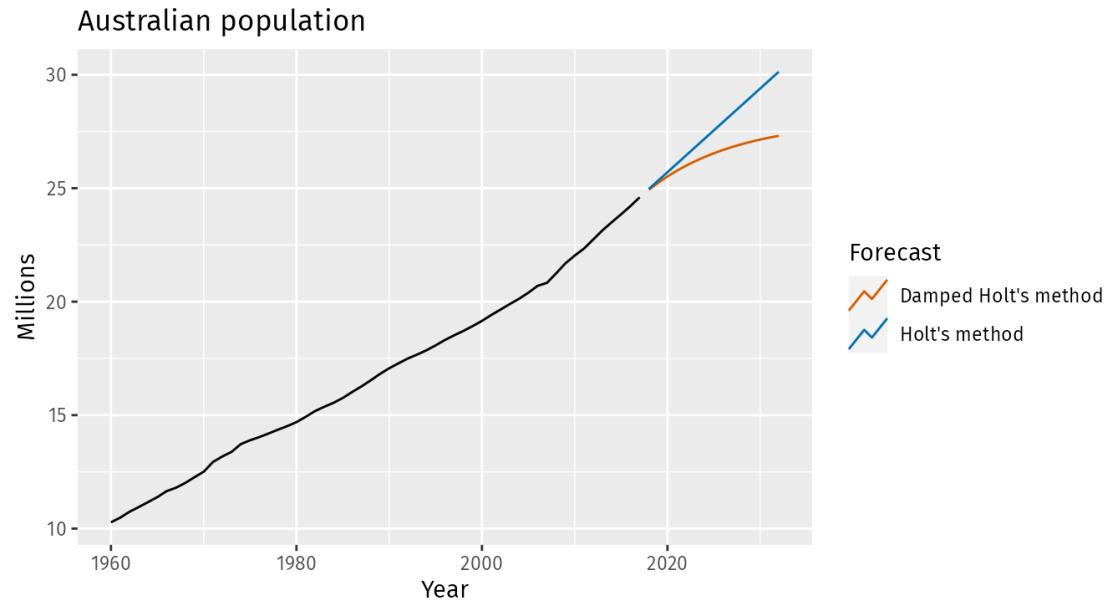
SES: Optimal Parameters*

Initial Level		1249.32				
Smoothing constant		0.61991			α	RMSE
t	Demand	Level	Forecast			68.85
0		1249.32			0.10	71.79
1	1212	1226.18	1249.32		0.20	72.27
2	1321	1284.96	1226.18		0.30	71.52
3	1278	1280.65	1284.96		0.40	70.35
4	1341	1318.06	1280.65		0.50	69.35
5	1257	1280.21	1318.06		0.60	68.87
6	1341	1317.89	1280.21		0.70	69.10
7	1257	1280.15	1317.89		0.80	70.18
8	1287	1284.39	1280.15		0.90	72.23
9	1189	1225.26	1284.39			
10	1111	1154.43	1225.26			
		MSE	4740.785			
		RMSE	68.85336			

- We need to select values of α and ℓ_0
- We find the values of the unknown parameters and the initial values that minimize MSE (or equivalently RMSE)

$$\text{Min}\{\text{RMSE}(\alpha): \alpha \in [0,1]\}$$

Simple exponential smoothing (SES)			
Initial Level (ℓ_0)		1212	
Smoothing parameter (α between 0,1)		0.695556943	
Time	Actual (y_t)	Level (ℓ_t)	Forecast
0		1212.000	
1	1212	1212.000	1,212.00
2	1321	1287.816	1,212.00
3	1278	1280.988	1,287.82
4	1341	1322.730	1,280.99
5	1257	1277.011	1,322.73
6	1341	1321.519	1,277.01
7	1257	1276.642	1,321.52
8	1287	1283.847	1,276.64
9	1189	1217.875	1,283.85
10	1111	1143.537	1,217.88
RMSE on train			69.77248



Methods with trend

Linear trend: Holt's method

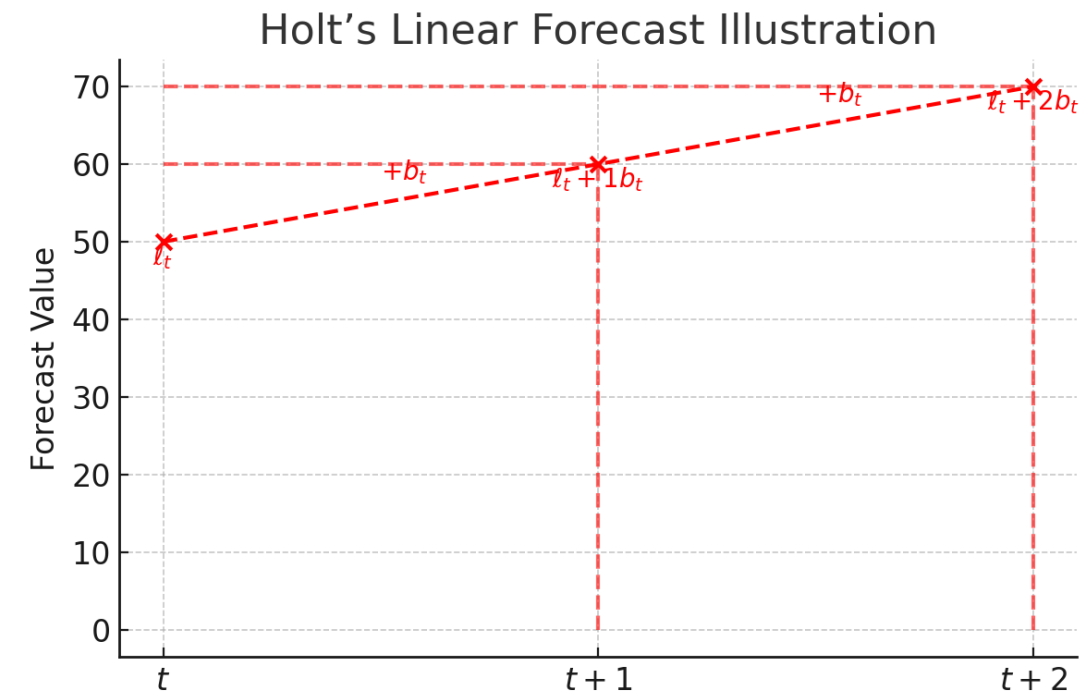
Damped trend

Holt's linear trend

Component form of Holt's linear trend

- Forecast: $\hat{y}_{t+h|t} = \ell_t + hb_t$

“Holt's FC = current baseline level + (steps ahead) × slope”



Holt's linear trend vs SES

- Recall component form of simple exponential smoothing

- Forecast: $\hat{y}_{t+h|t} =$

- Level: $\ell_t =$

weighted average b/w observation at time t & one-step forecast for time t

- Component form of Holt's linear trend

- Forecast: $\hat{y}_{t+h|t} = \ell_t + hb_t$

- Level: $\ell_t = \alpha y_t + (1 - \alpha)(\ell_{t-1} + b_{t-1})$

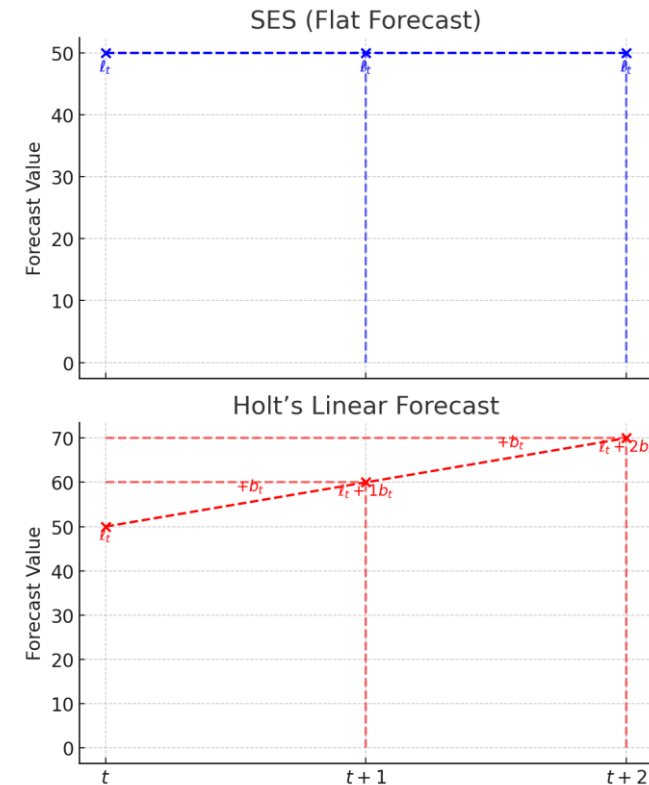
weighted average between y_t and one-step forecast for time t , $\hat{y}_{t|t-1}$

- Trend: $b_t = \beta^*(\ell_t - \ell_{t-1}) + (1 - \beta^*)b_{t-1}$

weighted average between $\ell_t - \ell_{t-1}$ and previous estimate of slope

T\S	N
	$\hat{y}_{t+h t} = \ell_t$
N	$\ell_t = \alpha y_t + (1 - \alpha)\ell_{t-1}$
	$\hat{y}_{t+h t} = \ell_t + hb_t$
A	$\ell_t = \alpha y_t + (1 - \alpha)(\ell_{t-1} + b_{t-1})$ $b_t = \beta^*(\ell_t - \ell_{t-1}) + (1 - \beta^*)b_{t-1}$

SES vs Holt's Linear Forecast



Holt's linear trend: Smoothing parameters

- Component form
 - Forecast: $\hat{y}_{t+h|t} = \ell_t + hb_t$
 - Level: $\ell_t = \alpha y_t + (1 - \alpha)(\ell_{t-1} + b_{t-1})$
 - Trend: $b_t = \beta^*(\ell_t - \ell_{t-1}) + (1 - \beta^*)b_{t-1}$
- Two smoothing parameters α and β^* where $0 \leq \alpha, \beta^* \leq 1$
- Choose $\alpha, \beta^*, \ell_0, b_0$ to minimize MSE



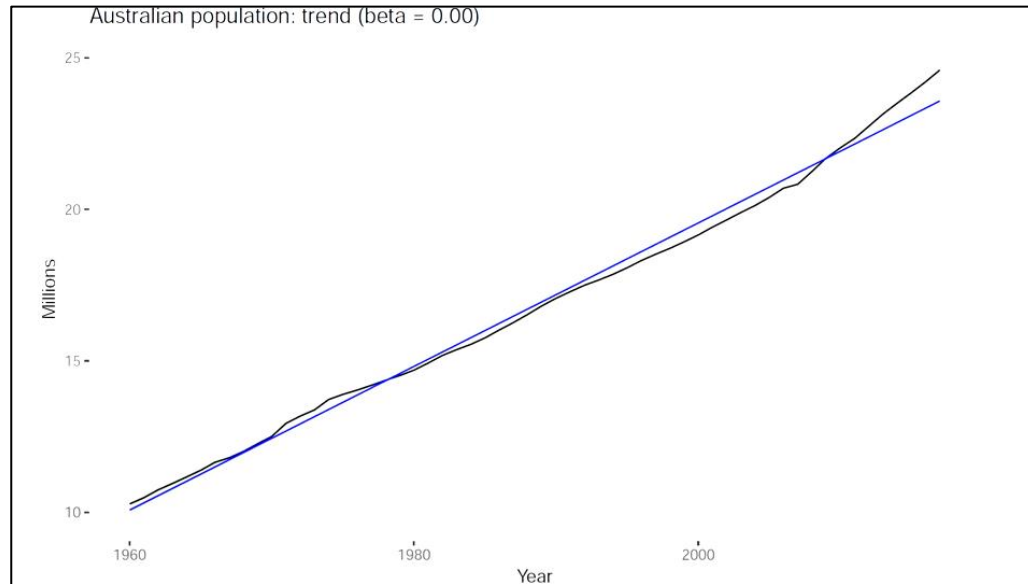
Holt's Linear Trend Method: Component Form

$$\begin{aligned} \text{Forecast equation} \quad & \hat{y}_{t+h|t} = \ell_t + hb_t \\ \text{Level equation} \quad & \ell_t = \alpha y_t + (1 - \alpha)(\ell_{t-1} + b_{t-1}) \\ \text{Trend equation} \quad & b_t = \beta^*(\ell_t - \ell_{t-1}) + (1 - \beta^*)b_{t-1}, \end{aligned}$$

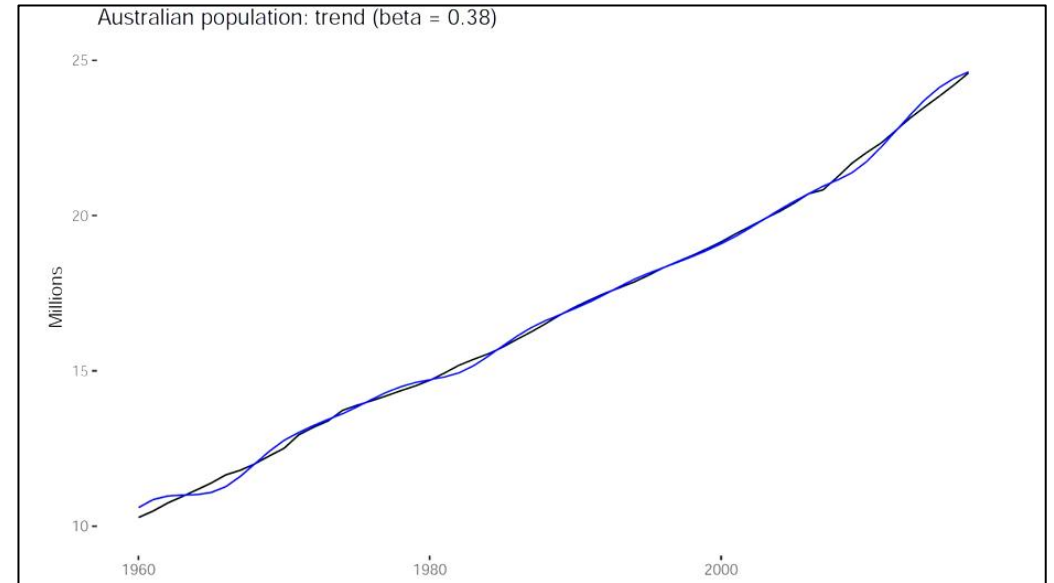
Initial		1324.771	-11.7524
Smoothing parameters		0.0001	0.0001
	t	Demand	Trend Forecast
	0	1324.771	-11.7524
	1	1212	1313.009
	2	1321	1301.258
	3	1278	1289.505
	4	1341	1277.759
	5	1257	1266.005
	6	1341	1254.262
	7	1257	1242.511
	8	1287	1230.764
	9	1189	1219.009
	10	1111	1207.247
	11		1,195.494
	12		1,183.742
	13		1,171.989
	14		1,160.237

Obtain 15-step forecast

Effect of the trend smoothing parameter β



Common misunderstanding: $\beta = 0$ does not remove the trend. It freezes the trend at its initial estimate.

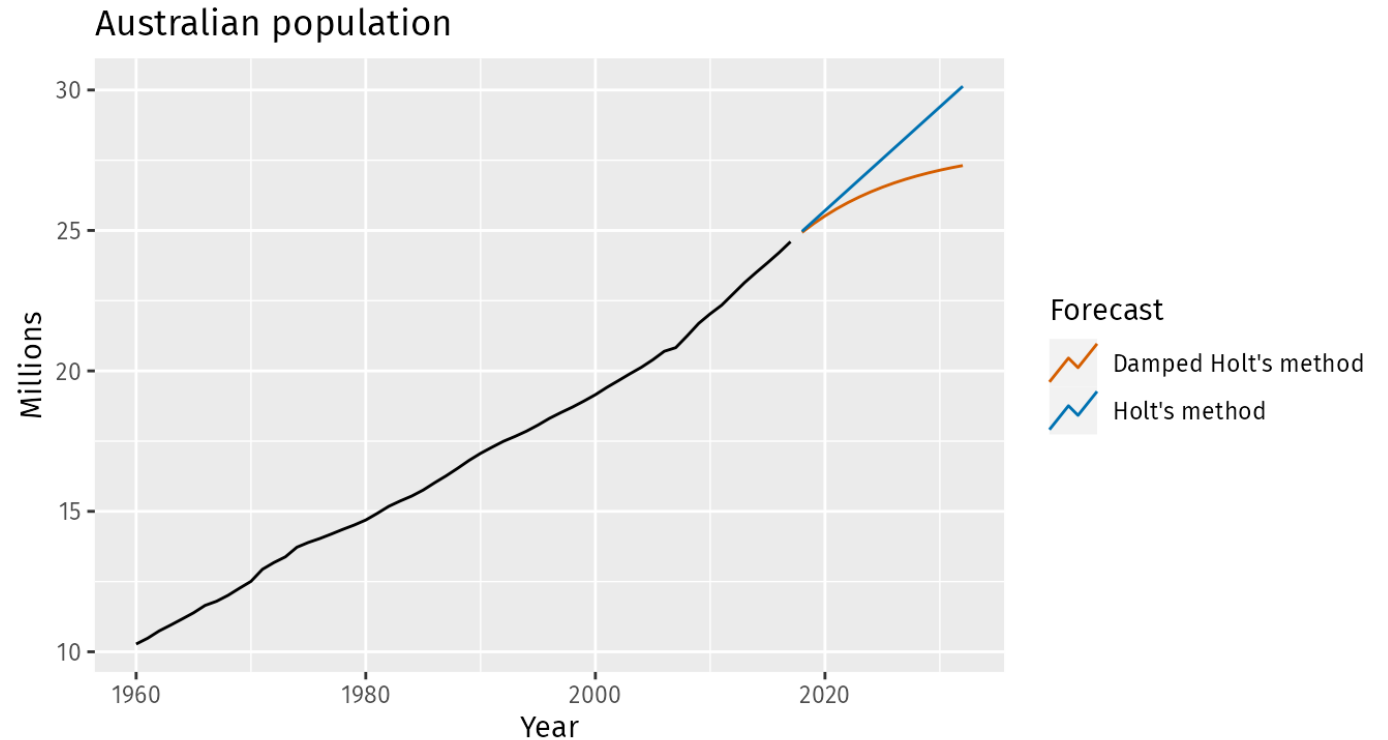


<https://youtu.be/sQir7ctNpVQ>

03:30 – 04:37

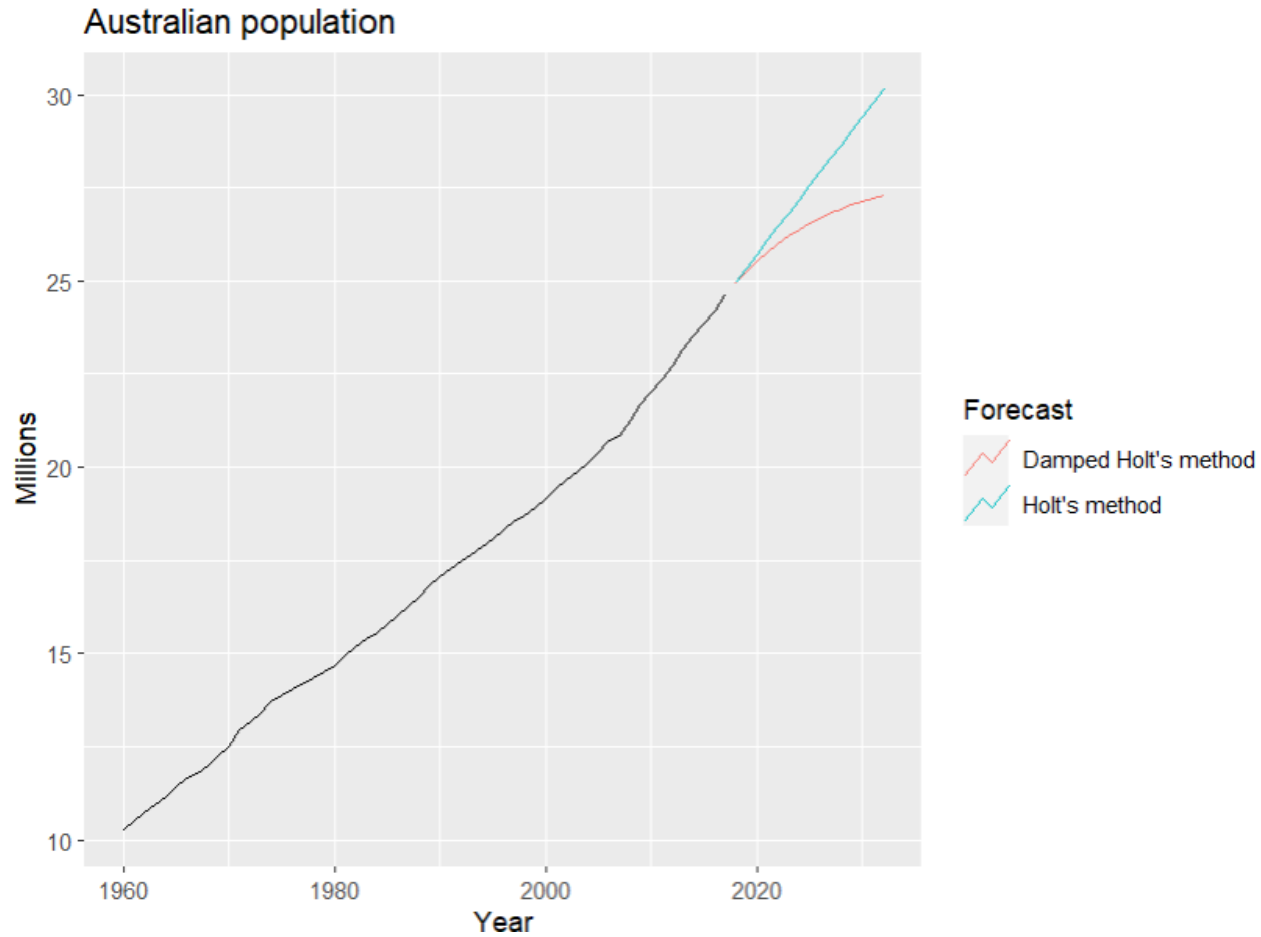
Damped trend methods

- Holt's linear method display a constant trend (increasing or decreasing) indefinitely into the future.
- Gardner and McKenzie (1985) introduced a parameter than "dampens" the trend to a flat line some time in future.
- Methods that include a damped trend have proven to be very successful, and are arguably the most popular individual methods when forecasts are required automatically for many series.

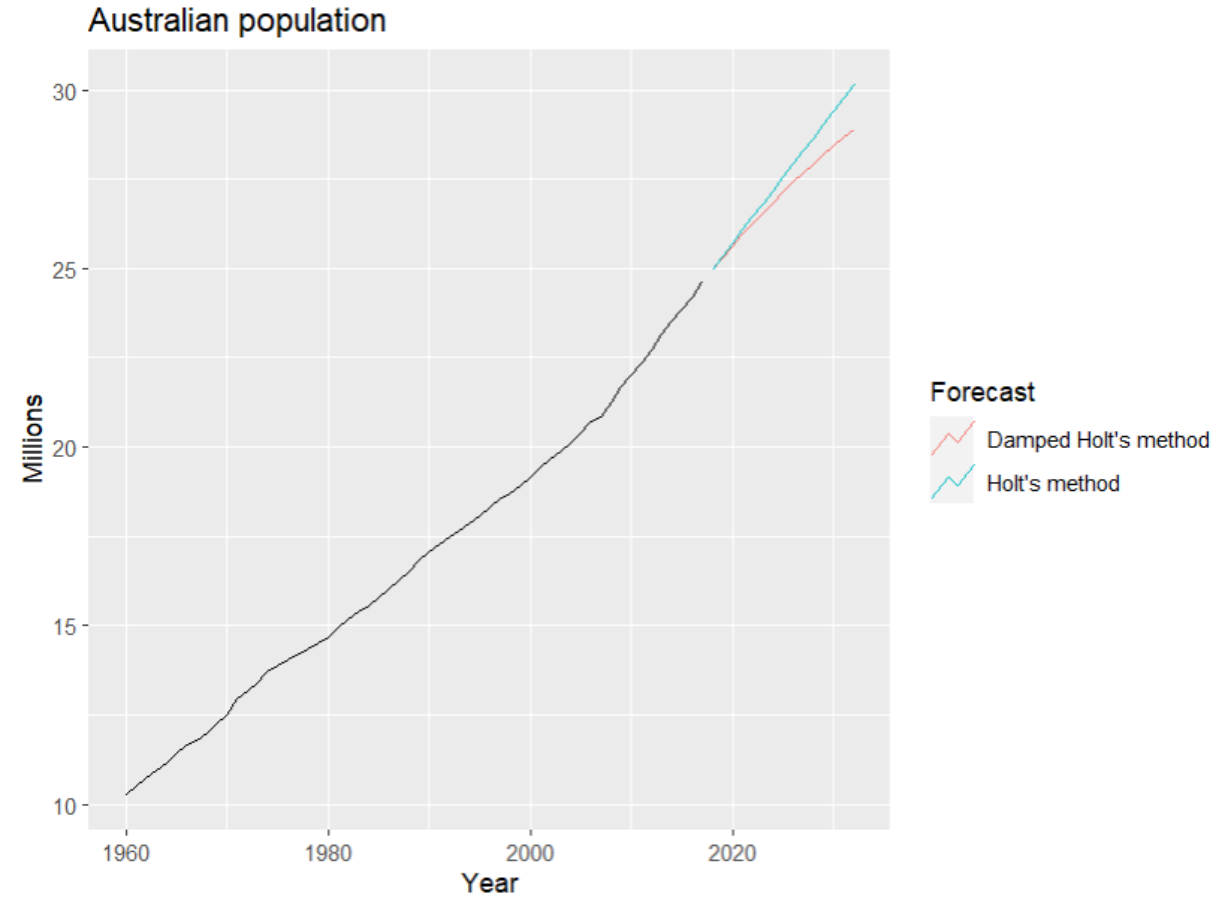


```
aus_economy |>
  model(
    `Holt's method` = ETS(Pop ~ error("A") +
                          trend("A") + season("N")),
    `Damped Holt's method` = ETS(Pop ~ error("A") +
                                trend("Ad", phi = 0.9) + season("N"))
  ) |>
  forecast(h = 15) |>
  autoplot(aus_economy, level = NULL) +
  labs(title = "Australian population",
       y = "Millions") +
  guides(colour = guide_legend(title = "Forecast"))
```

Damping parameter



$$\phi = 0.95$$



$$\phi = 0.97$$

Damped trend methods

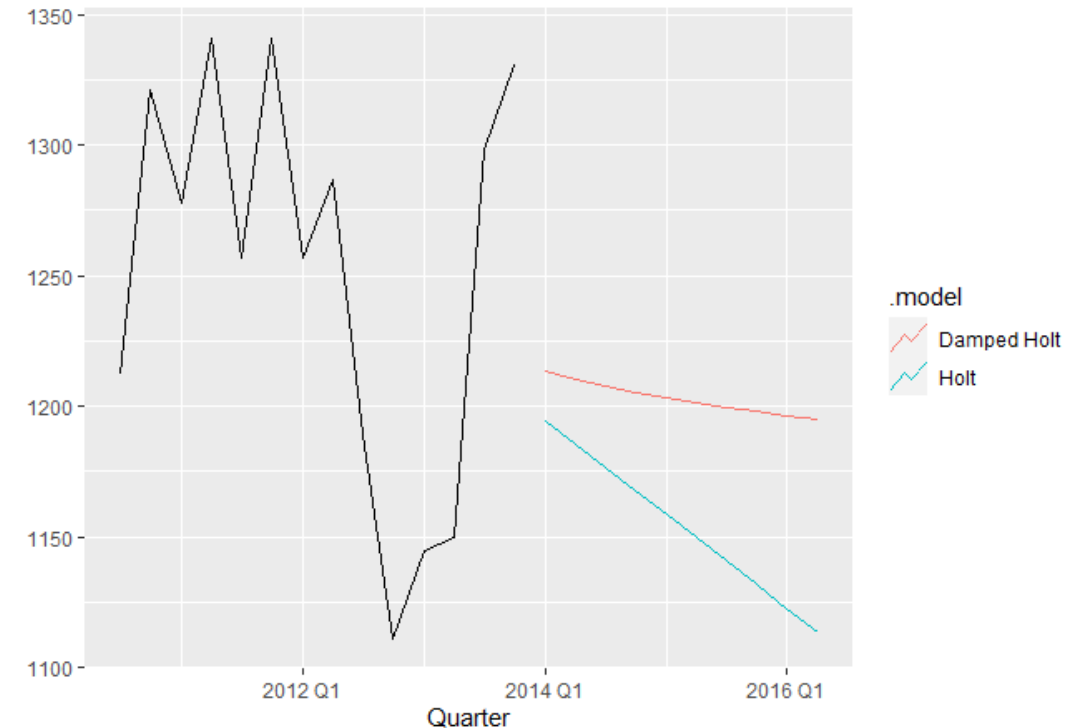
Holt's linear method display a constant trend (increasing or decreasing) indefinitely into the future.

Gardner and McKenzie (1985) introduced a parameter than "dampens" the trend to a flat line some time in future.

- Let $\phi \in (0,1]$ be the **damping parameter**.

$$\begin{aligned}\hat{y}_{t+h|t} &= \ell_t + (\phi + \phi^2 + \dots + \phi^h)b_t \\ \ell_t &= \alpha y_t + (1 - \alpha)(\ell_{t-1} + \phi b_{t-1}) \\ b_t &= \beta^*(\ell_t - \ell_{t-1}) + (1 - \beta^*)\phi b_{t-1}.\end{aligned}$$

- In practice, $\phi \in [0.80, 0.98]$.
- If $\phi = 1$, this method is identical to Holt's linear trend. Trend continue forever.
- If $\phi = 0$, this method is identical to SES. No trend.



```
y %>% model(  
  "Holt" = ETS(Obs ~ error("A") + trend("A") + season("N")),  
  "Damped Holt" = ETS(Obs ~ error("A") + trend("Ad", phi = 0.9) + season("N"))  
) %>%  
  forecast(h=10) %>%  
  autoplot(y, level = NULL)
```

Add to the plot, forecast for $\phi = 0.95$

Illustration next slide



Damped trend methods

Sheet: selectedETS

$$\hat{y}_{t+h|t} = l_t + (\phi + \phi^2 + \dots + \phi^h)b_t$$

$$l_t = \alpha y_t + (1 - \alpha)(l_{t-1} + \phi b_{t-1})$$

$$b_t = \beta^*(l_t - l_{t-1}) + (1 - \beta^*)\phi b_{t-1}.$$

Illustration of various damping parameters

- If $\phi = 1$, this method is identical to Holt's linear trend. Trend continue forever.
- If $\phi = 0$, this method is identical to SES. No trend.

Holt's damped trend						
Damping parameter ϕ		0.9				
Initial l_0, b_0		1324.771	-11.7524	What if $\phi = 1.0$?		
Smoothing parameters		0.0001	0.0001			
Time period	Actual (y_t)	Level (l_t)	Trend (b_t)	Forecast	ϕ^h	CumSum ϕ^h
	0	1324.771	-11.752400			
	1	1212	1314.184	-10.577161		
	2	1321	1304.666	-9.519445		
	3	1278	1296.096	-8.567500		
	4	1341	1288.391	-7.710750		
	5	1257	1281.449	-6.939675		
	6	1341	1275.21	-6.245707		
	7	1257	1269.587	-5.621136		
	8	1287	1264.531	-5.059023		
	9	1189	1259.97	-4.553121		
h	10	1111	1255.858	-4.097810		
1	11			1,252.17	0.9000	0.9000
2	12			1,248.85	0.8100	1.7100
3	13			1,245.86	0.7290	2.4390
4	14			1,243.17	0.6561	3.0951
5	15			1,240.76	0.5905	3.6856
6	16			1,238.58	0.5314	4.2170
7	17			1,236.62	0.4783	4.6953
8	18			1,234.85	0.4305	5.1258
9	19			1,233.27	0.3874	5.5132
10	20			1,231.84	0.3487	5.8619
11	21			1,230.55	0.3138	6.1757
12	22			1,229.39	0.2824	6.4581
13	23			1,228.35	0.2542	6.7123

Example: Internet usage

Interpreting parameters

```
fit <- www_usage %>%  
  model(  
    Damped = ETS(value ~ error("A") + trend("Ad") +  
                  season("N"))  
  )
```

Estimated parameters:

```
tidy(fit)
```

```
fit %>%
```

```
  forecast(h = 10) %>%
```

```
  autoplot(www_usage) +
```

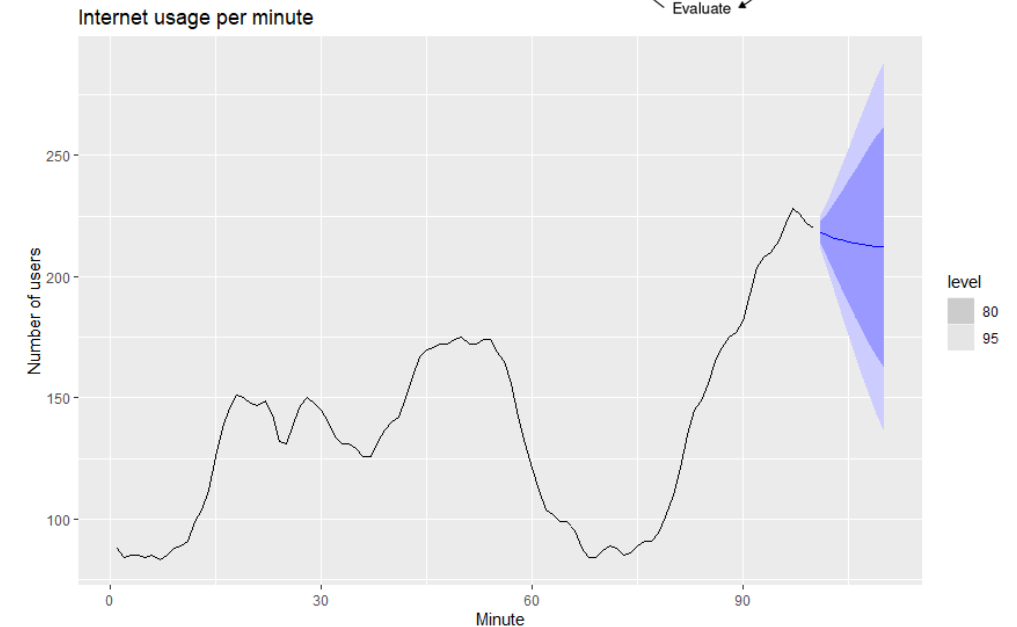
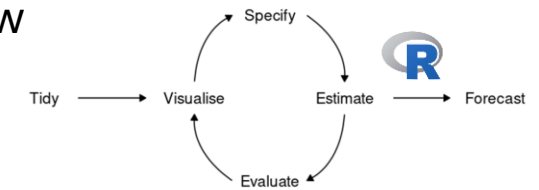
```
  labs(x="Minute", y="Number of users",
```

```
        title = "Internet usage per minute")
```

```
# A tibble: 5 x 3  
  .model term estimate  
  <chr>   <chr>   <dbl>  
1 Damped alpha  1.00  
2 Damped beta   0.997  
3 Damped phi    0.815  
4 Damped l[0]   90.4  
5 Damped b[0]  -0.0173
```

The resulting forecasts look sensible with decreasing trend, which flattens out due to the low value of the damping parameter (0.815), and relatively wide prediction intervals reflecting the variation in the historical data.

Forecasting flow



 R automatically estimates the smoothing parameters (α, β, ϕ) and initial states (ℓ_0, b_0) for us.



We don't have to hand-tune them. Instead, we can focus on interpreting the parameters.

- $\alpha \approx 1$, showing that the level reacts strongly to each new observation.
- $\beta \approx 1$: the trend changes to mostly reflect the slope between the last two minutes of internet usage.
- $\phi = 0.815$: Trend is damped and flatten out over time.

3. งานศึกษาด้านเศรษฐกิจและสังคม

Forecast: Province level

Innovation state space model for exponential smoothing ETS(A,Ad,N): Damped trend method

$$\begin{aligned}\hat{y}_{t+h|t} &= \ell_t + (\phi + \phi^2 + \dots + \phi^h)b_t \\ \ell_t &= \alpha y_t + (1 - \alpha)(\ell_{t-1} + \phi b_{t-1}) \\ b_t &= \beta^*(\ell_t - \ell_{t-1}) + (1 - \beta^*)\phi b_{t-1}\end{aligned}$$

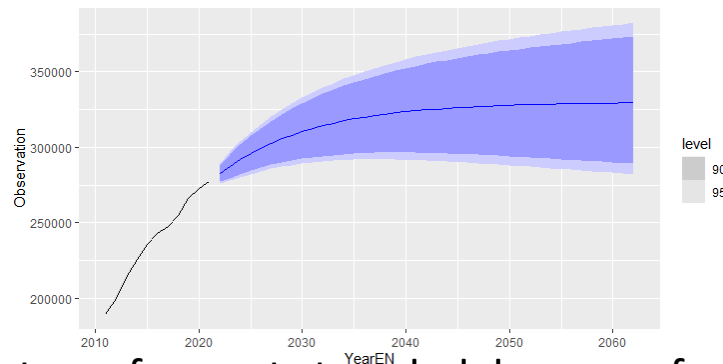
```
# forecast using ETS after log transform
fit <- doh %>%
  model(ets = ETS(log(Observation) ~ error("A") + trend("Ad") + season("N")))

# obtain forecast fable
fc <- fit %>%
  forecast(h= 50) %>%      # 50-year-ahead FC. Make sure this covers yearEnd
  filter(YearEN <= yrEnd)

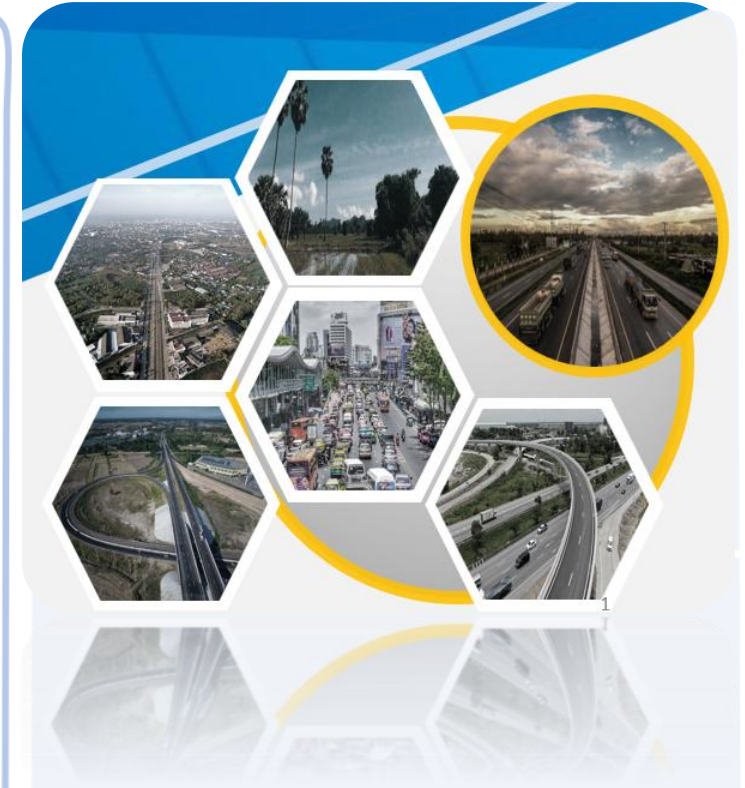
# Progress slide: report model for selected variable and each province
fit %>% |
  filter(Type == "Household", ProvinceID == "83") %>%
  tidy()
# plot of point forecast and prediction intervals
fc %>%
  filter(Type == "Household", ProvinceID == "83") %>%
  autoplot(doh, level = c(90,95))
```



term	estimate
<chr>	<dbl>
alpha	1.00
beta	0.00375
phi	0.887
l[0]	12.1
b[0]	0.0816



- As $h \rightarrow \infty$, $\hat{y}_{t+h|t} \rightarrow \ell_t + \frac{\phi}{1-\phi} b_t$. Short-run forecasts trended, long-run forecasts constant.



- k is the integer part of $(h - 1)/m$

Trend	Seasonal		
	N	A	M
N	$\hat{y}_{t+h t} = \ell_t$ $\ell_t = \alpha y_t + (1 - \alpha)\ell_{t-1}$	$\hat{y}_{t+h t} = \ell_t + s_{t+h-m(k+1)}$ $\ell_t = \alpha(y_t - s_{t-m}) + (1 - \alpha)\ell_{t-1}$ $s_t = \gamma(y_t - \ell_{t-1}) + (1 - \gamma)s_{t-m}$	$\hat{y}_{t+h t} = \ell_t s_{t+h-m(k+1)}$ $\ell_t = \alpha(y_t/s_{t-m}) + (1 - \alpha)\ell_{t-1}$ $s_t = \gamma(y_t/\ell_{t-1}) + (1 - \gamma)s_{t-m}$
A	$\hat{y}_{t+h t} = \ell_t + hb_t$ $\ell_t = \alpha y_t + (1 - \alpha)(\ell_{t-1} + b_{t-1})$ $b_t = \beta^*(\ell_t - \ell_{t-1}) + (1 - \beta^*)b_{t-1}$	$\hat{y}_{t+h t} = \ell_t + hb_t + s_{t+h-m(k+1)}$ $\ell_t = \alpha(y_t - s_{t-m}) + (1 - \alpha)(\ell_{t-1} + b_{t-1})$ $b_t = \beta^*(\ell_t - \ell_{t-1}) + (1 - \beta^*)b_{t-1}$ $s_t = \gamma(y_t - \ell_{t-1} - b_{t-1}) + (1 - \gamma)s_{t-m}$	$\hat{y}_{t+h t} = (\ell_t + hb_t)s_{t+h-m(k+1)}$ $\ell_t = \alpha(y_t/s_{t-m}) + (1 - \alpha)(\ell_{t-1} + b_{t-1})$ $b_t = \beta^*(\ell_t - \ell_{t-1}) + (1 - \beta^*)b_{t-1}$ $s_t = \gamma(y_t/(\ell_{t-1} + b_{t-1})) + (1 - \gamma)s_{t-m}$
A _d	$\hat{y}_{t+h t} = \ell_t + \phi_h b_t$ $\ell_t = \alpha y_t + (1 - \alpha)(\ell_{t-1} + \phi b_{t-1})$ $b_t = \beta^*(\ell_t - \ell_{t-1}) + (1 - \beta^*)\phi b_{t-1}$	$\hat{y}_{t+h t} = \ell_t + \phi_h b_t + s_{t+h-m(k+1)}$ $\ell_t = \alpha(y_t - s_{t-m}) + (1 - \alpha)(\ell_{t-1} + \phi b_{t-1})$ $b_t = \beta^*(\ell_t - \ell_{t-1}) + (1 - \beta^*)\phi b_{t-1}$ $s_t = \gamma(y_t - \ell_{t-1} - \phi b_{t-1}) + (1 - \gamma)s_{t-m}$	$\hat{y}_{t+h t} = (\ell_t + \phi_h b_t)s_{t+h-m(k+1)}$ $\ell_t = \alpha(y_t/s_{t-m}) + (1 - \alpha)(\ell_{t-1} + \phi b_{t-1})$ $b_t = \beta^*(\ell_t - \ell_{t-1}) + (1 - \beta^*)\phi b_{t-1}$ $s_t = \gamma(y_t/(\ell_{t-1} + \phi b_{t-1})) + (1 - \gamma)s_{t-m}$

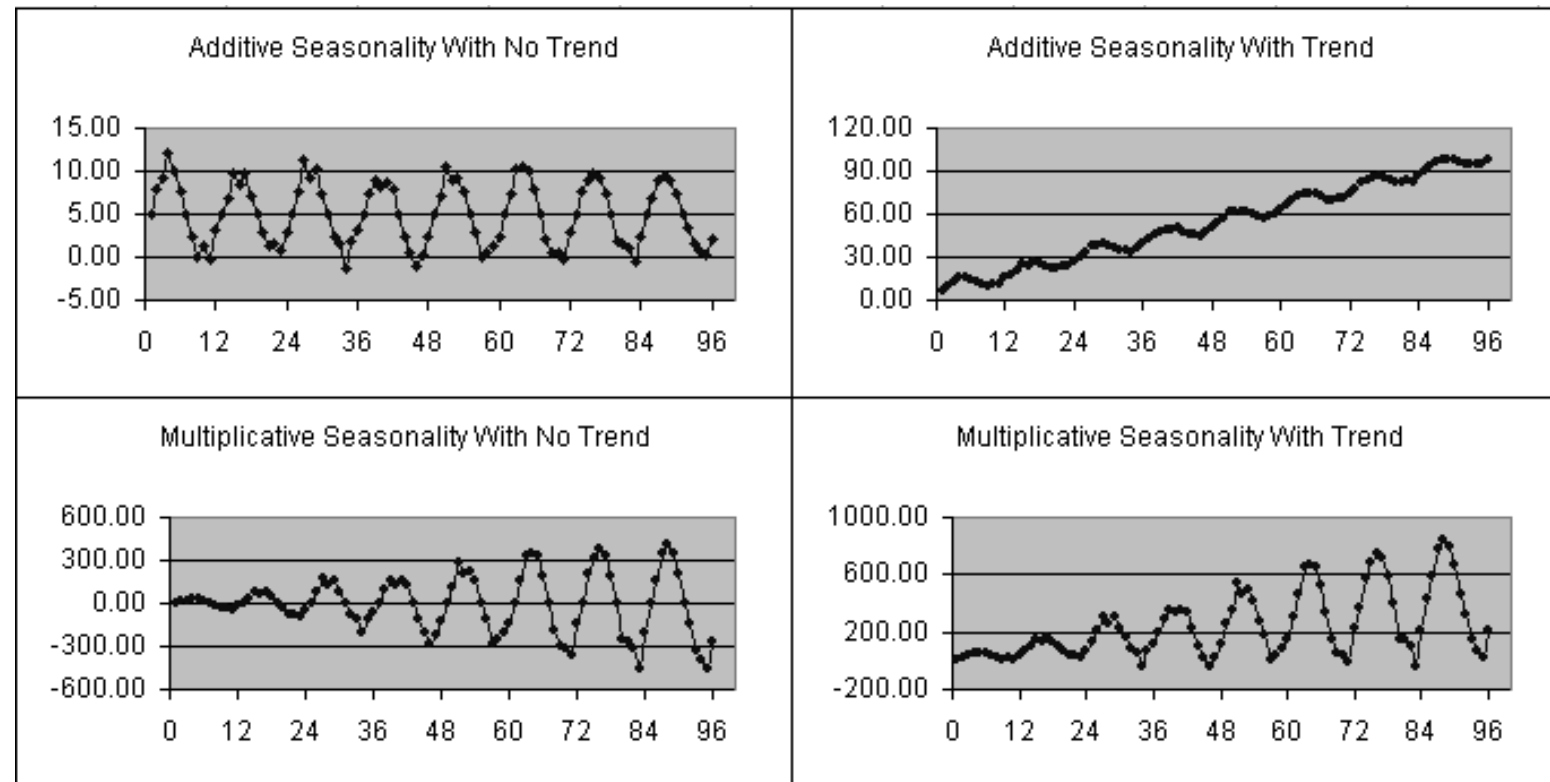
- $\phi_h = \phi + \phi^2 + \phi^3 + \dots + \phi^h$
- If $\phi = 1$, identical to Holt's linear trend.
- As $h \rightarrow \infty$, $\hat{y}_{t+h|t} \rightarrow \ell_t + \frac{\phi}{1-\phi} b_t$. Short-run forecasts trended, long-run forecasts constant.

Methods with seasonality

Trend Component	Seasonal Component		
	N	A	M
	(None)	(Additive)	(Multiplicative)
N (None)	(N,N)	(N,A)	(N,M)
A (Additive)	(A,N)	(A,A)	(A,M)
A_d (Additive damped)	(A_d ,N)	(A_d ,A)	(A_d ,M)

Table 8.5: A two-way classification of exponential smoothing methods.

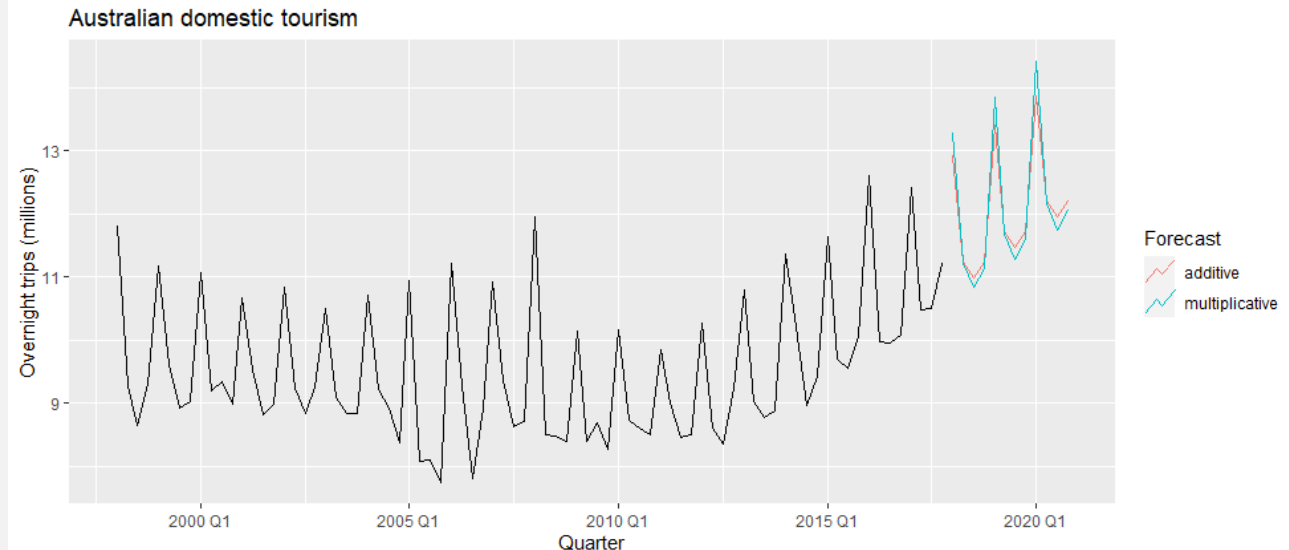
Trend Component	Seasonal Component		
	N	A	M
	(None)	(Additive)	(Multiplicative)
N (None)	(N,N)	(N,A)	(N,M)
A (Additive)	(A,N)	(A,A)	(A,M)
A_d (Additive damped)	(A_d ,N)	(A_d ,A)	(A_d ,M)



Example: Domestic overnight trips in Australia

```
aus_holidays <- tourism %>%  
  filter(Purpose == "Holiday") %>%  
  summarise(Trips = sum(Trips)/1e3)  
  
fit <- aus_holidays %>%  
  model(  
    additive = ETS(Trips ~ error("A") + trend("A") +  
      season("A")),  
    multiplicative = ETS(Trips ~ error("M") + trend("A") +  
      season("M"))  
  )  
  
fc <- fit %>% forecast(h = "3 years")  
fc %>%  
  autoplot(aus_holidays, level = NULL) +  
  labs(title="Australian domestic tourism",  
    y="Overnight trips (millions)") +  
  guides(colour = guide_legend(title = "Forecast"))
```

```
> aus_holidays  
# A tsibble: 80 x 2 [1Q]  
  Quarter Trips  
  <qtr> <dbl>  
1 1998 Q1 11.8  
2 1998 Q2 9.28  
3 1998 Q3 8.64  
4 1998 Q4 9.30  
5 1999 Q1 11.2  
6 1999 Q2 9.61  
7 1999 Q3 8.91  
8 1999 Q4 9.03  
9 2000 Q1 11.1  
10 2000 Q2 9.20  
# ... with 70 more rows
```



Seasonal indices

- The additive seasonal index sums to approximately zero.
- The multiplicative seasonal index sums to approximately $m = 4$ for quarterly data

```
> aus_holidays
# A tsibble: 80 x 2 [1Q]
  Quarter Trips
  <qtr> <dbl>
1 1998 Q1 11.8
2 1998 Q2 9.28
3 1998 Q3 8.64
4 1998 Q4 9.30
5 1999 Q1 11.2
6 1999 Q2 9.61
7 1999 Q3 8.91
8 1999 Q4 9.03
9 2000 Q1 11.1
10 2000 Q2 9.20
# ... with 70 more rows
```

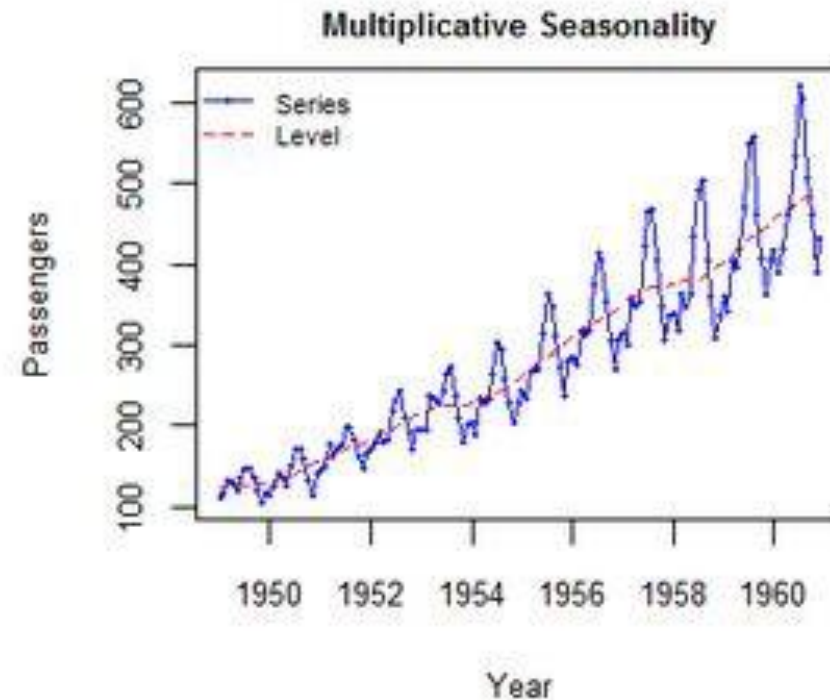
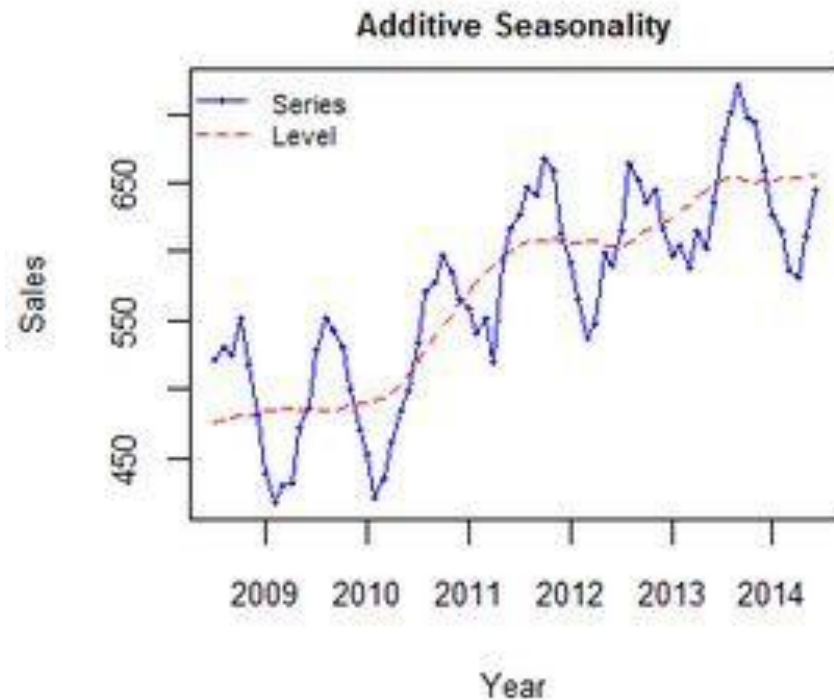
```
> tail(aus_holidays)
# A tsibble: 6 x 2 [1Q]
  Quarter Trips
  <qtr> <dbl>
1 2016 Q3 9.94
2 2016 Q4 10.1
3 2017 Q1 12.4
4 2017 Q2 10.5
5 2017 Q3 10.5
6 2017 Q4 11.2
```

```
> tidy(fit)
# A tibble: 18 x 3
  .model term estimate
  <chr> <chr> <dbl>
1 additive alpha 0.262
2 additive beta 0.0431
3 additive gamma 0.000100
4 additive l[0] 9.79
5 additive b[0] 0.0211
6 additive s[0] -0.534 Q4
7 additive s[-1] -0.670 Q3
8 additive s[-2] -0.294 Q2
9 additive s[-3] 1.50 Q1
10 multiplicative alpha 0.224
11 multiplicative beta 0.0304
12 multiplicative gamma 0.000100
13 multiplicative l[0] 10.0
14 multiplicative b[0] -0.0114
15 multiplicative s[0] 0.943
16 multiplicative s[-1] 0.927
17 multiplicative s[-2] 0.969
18 multiplicative s[-3] 1.16
```

Additive vs Multiplicative seasonality

$$\hat{y}_{t+h|t} = \ell_t + hb_t + s_{t+h-m(k+1)}$$

$$\hat{y}_{t+h|t} = (\ell_t + hb_t)s_{t+h-m(k+1)}$$



Suppose that the non-seasonal forecast is 400.

Additive: Suppose seasonal index of Q2 is 40.
Then, seasonal fc is

Multiplicative: Suppose seasonal index of Q2 is 1.40.

Repeat the calculation for the non-seasonal forecast of 500.

Holt-Winters' Seasonal Method

- m = frequency of the seasonality, e.g., quarterly data $m = 4$, monthly data $m = 12$
- k = integer part of $(h - 1)/m$, which ensures that the estimates of the seasonal indices used for forecasting come from final year of the sample.

HW Additive

$$\begin{aligned}\hat{y}_{t+h|t} &= \ell_t + hb_t + s_{t+h-m(k+1)} \\ \ell_t &= \alpha(y_t - s_{t-m}) + (1 - \alpha)(\ell_{t-1} + b_{t-1}) \\ b_t &= \beta^*(\ell_t - \ell_{t-1}) + (1 - \beta^*)b_{t-1} \\ s_t &= \gamma(y_t - \ell_{t-1} - b_{t-1}) + (1 - \gamma)s_{t-m},\end{aligned}$$

- Additive method is preferred when the seasonal variations are roughly constant through the series.

HW Multiplicative

$$\begin{aligned}\hat{y}_{t+h|t} &= (\ell_t + hb_t)s_{t+h-m(k+1)} \\ \ell_t &= \alpha \frac{y_t}{s_{t-m}} + (1 - \alpha)(\ell_{t-1} + b_{t-1}) \\ b_t &= \beta^*(\ell_t - \ell_{t-1}) + (1 - \beta^*)b_{t-1} \\ s_t &= \gamma \frac{y_t}{(\ell_{t-1} + b_{t-1})} + (1 - \gamma)s_{t-m}\end{aligned}$$

- Multiplicative method is preferred when the seasonal variations are changing proportional to the level of the series

Holt-Winters' damped method

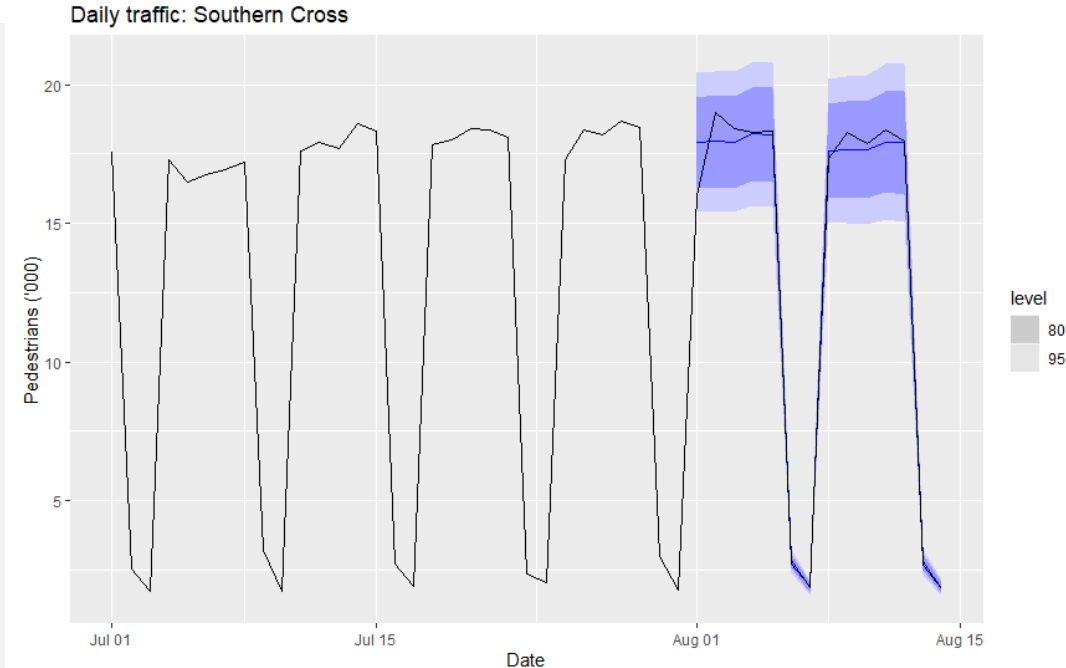
- Damping is possible with both additive and multiplicative Holt-Winters' methods.
- A method that often provides accurate and robust forecasts for seasonal data is the Holt-Winters method with a damped trend and multiplicative seasonality:

$$\begin{aligned}\hat{y}_{t+h|t} &= [\ell_t + (\phi + \phi^2 + \dots + \phi^h)b_t] s_{t+h-m(k+1)} \\ \ell_t &= \alpha(y_t/s_{t-m}) + (1 - \alpha)(\ell_{t-1} + \phi b_{t-1}) \\ b_t &= \beta^*(\ell_t - \ell_{t-1}) + (1 - \beta^*)\phi b_{t-1} \\ s_t &= \gamma \frac{y_t}{(\ell_{t-1} + \phi b_{t-1})} + (1 - \gamma)s_{t-m}.\end{aligned}$$

Example: Holt-Winters method with daily data

Holt-Winters method with a damped trend and multiplicative seasonality

```
sth_cross_ped <- pedestrian %>%
  filter(Date >= "2016-07-01",
         Sensor == "Southern Cross Station") %>%
  index_by(Date) %>%
  summarise(Count = sum(Count)/1000)
sth_cross_ped %>%
  filter(Date <= "2016-07-31") %>%
  model(
    hw = ETS(Count ~ error("M") + trend("Ad") + season("M"))
  ) %>%
  forecast(h = "2 weeks") %>%
  autoplot(sth_cross_ped %>% filter(Date <= "2016-08-14")) +
  labs(title = "Daily traffic: Southern Cross",
       y="Pedestrians ('000)")
```



Daily data: seasonal period $m = 7$
Weekly seasonality can be declared explicitly:

```
season("A", period = 7)
```

The appropriate unit of time for h is in days.

ETS methods vs models

State space model

We label each state space model as

ETS($\cdot, \cdot, \cdot$) for (**Error**, **Trend**, **Seasonal**)

Error = {A, M}

Trend = {N, A, A_d }

Seasonal = {N, A, M}

Additive Error		Seasonal Component		
	Trend Component	N	A	M
		(None)	(Additive)	(Multiplicative)
N	(None)	A, N, N	A, N, A	A, N, M
A	(Additive)	A, A, N	A, A, A	A, A, M
A_d	(Additive damped)	A, A_d , N	A, A_d , A	A, A_d , M

Multiplicative Error		Seasonal Component		
	Trend Component	N	A	M
		(None)	(Additive)	(Multiplicative)
N	(None)	M, N, N	M, N, A	M, N, M
A	(Additive)	M, A, N	M, A, A	M, A, M
A_d	(Additive damped)	M, A_d , N	M, A_d , A	M, A_d , M

Methods

Trend	Seasonal		
	N	A	M
N	$\hat{y}_{t+h t} = \ell_t$	$\hat{y}_{t+h t} = \ell_t + s_{t+h-m(k+1)}$	$\hat{y}_{t+h t} = \ell_t s_{t+h-m(k+1)}$
	$\ell_t = \alpha y_t + (1 - \alpha)\ell_{t-1}$	$\ell_t = \alpha(y_t - s_{t-m}) + (1 - \alpha)\ell_{t-1}$	$\ell_t = \alpha(y_t/s_{t-m}) + (1 - \alpha)\ell_{t-1}$
		$s_t = \gamma(y_t - \ell_{t-1}) + (1 - \gamma)s_{t-m}$	$s_t = \gamma(y_t/\ell_{t-1}) + (1 - \gamma)s_{t-m}$
A	$\hat{y}_{t+h t} = \ell_t + hb_t$	$\hat{y}_{t+h t} = \ell_t + hb_t + s_{t+h-m(k+1)}$	$\hat{y}_{t+h t} = (\ell_t + hb_t)s_{t+h-m(k+1)}$
	$\ell_t = \alpha y_t + (1 - \alpha)(\ell_{t-1} + b_{t-1})$	$\ell_t = \alpha(y_t - s_{t-m}) + (1 - \alpha)(\ell_{t-1} + b_{t-1})$	$\ell_t = \alpha(y_t/s_{t-m}) + (1 - \alpha)(\ell_{t-1} + b_{t-1})$
	$b_t = \beta^*(\ell_t - \ell_{t-1}) + (1 - \beta^*)b_{t-1}$	$b_t = \beta^*(\ell_t - \ell_{t-1}) + (1 - \beta^*)b_{t-1}$	$b_t = \beta^*(\ell_t - \ell_{t-1}) + (1 - \beta^*)b_{t-1}$
		$s_t = \gamma(y_t - \ell_{t-1} - b_{t-1}) + (1 - \gamma)s_{t-m}$	$s_t = \gamma(y_t/(\ell_{t-1} + b_{t-1})) + (1 - \gamma)s_{t-m}$
Ad	$\hat{y}_{t+h t} = \ell_t + \phi_h b_t$	$\hat{y}_{t+h t} = \ell_t + \phi_h b_t + s_{t+h-m(k+1)}$	$\hat{y}_{t+h t} = (\ell_t + \phi_h b_t)s_{t+h-m(k+1)}$
	$\ell_t = \alpha y_t + (1 - \alpha)(\ell_{t-1} + \phi b_{t-1})$	$\ell_t = \alpha(y_t - s_{t-m}) + (1 - \alpha)(\ell_{t-1} + \phi b_{t-1})$	$\ell_t = \alpha(y_t/s_{t-m}) + (1 - \alpha)(\ell_{t-1} + \phi b_{t-1})$
	$b_t = \beta^*(\ell_t - \ell_{t-1}) + (1 - \beta^*)\phi b_{t-1}$	$b_t = \beta^*(\ell_t - \ell_{t-1}) + (1 - \beta^*)\phi b_{t-1}$	$b_t = \beta^*(\ell_t - \ell_{t-1}) + (1 - \beta^*)\phi b_{t-1}$
		$s_t = \gamma(y_t - \ell_{t-1} - \phi b_{t-1}) + (1 - \gamma)s_{t-m}$	$s_t = \gamma(y_t/(\ell_{t-1} + \phi b_{t-1})) + (1 - \gamma)s_{t-m}$

Models

ADDITIVE ERROR MODELS

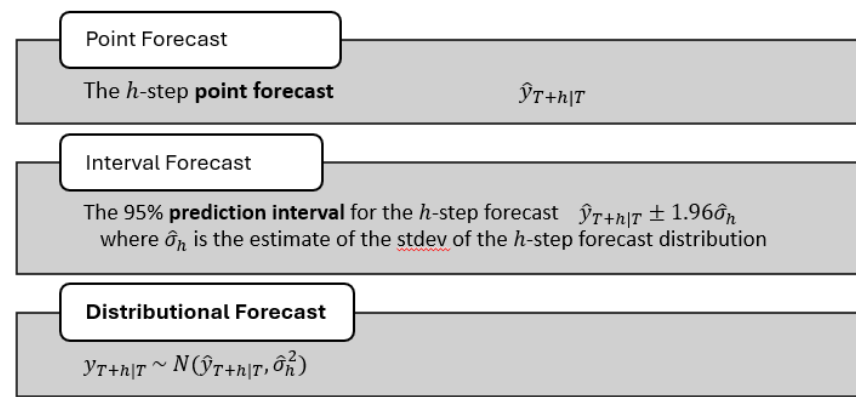
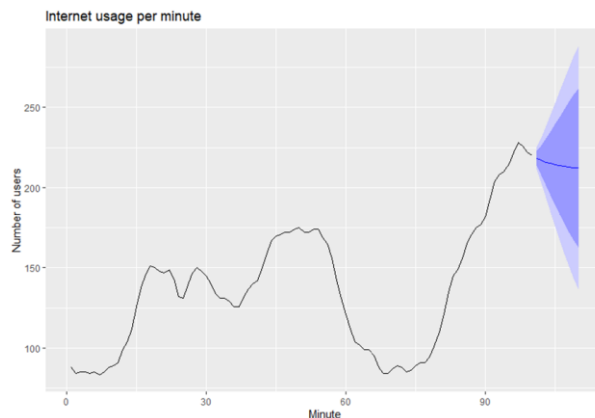
Trend	Seasonal		
	N	A	M
N	$y_t = \ell_{t-1} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \alpha \varepsilon_t$	$y_t = \ell_{t-1} + s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \alpha \varepsilon_t$ $s_t = s_{t-m} + \gamma \varepsilon_t$	$y_t = \ell_{t-1} s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \alpha \varepsilon_t / s_{t-m}$ $s_t = s_{t-m} + \gamma \varepsilon_t / \ell_{t-1}$
A	$y_t = \ell_{t-1} + b_{t-1} + \varepsilon_t$ $\ell_t = \ell_{t-1} + b_{t-1} + \alpha \varepsilon_t$ $b_t = b_{t-1} + \beta \varepsilon_t$	$y_t = \ell_{t-1} + b_{t-1} + s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + b_{t-1} + \alpha \varepsilon_t$ $b_t = b_{t-1} + \beta \varepsilon_t$ $s_t = s_{t-m} + \gamma \varepsilon_t$	$y_t = (\ell_{t-1} + b_{t-1})s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + b_{t-1} + \alpha \varepsilon_t / s_{t-m}$ $b_t = b_{t-1} + \beta \varepsilon_t / s_{t-m}$ $s_t = s_{t-m} + \gamma \varepsilon_t / (\ell_{t-1} + b_{t-1})$
Ad	$y_t = \ell_{t-1} + \phi b_{t-1} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \phi b_{t-1} + \alpha \varepsilon_t$ $b_t = \phi b_{t-1} + \beta \varepsilon_t$	$y_t = \ell_{t-1} + \phi b_{t-1} + s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \phi b_{t-1} + \alpha \varepsilon_t$ $b_t = \phi b_{t-1} + \beta \varepsilon_t$ $s_t = s_{t-m} + \gamma \varepsilon_t$	$y_t = (\ell_{t-1} + \phi b_{t-1})s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \phi b_{t-1} + \alpha \varepsilon_t / s_{t-m}$ $b_t = \phi b_{t-1} + \beta \varepsilon_t / s_{t-m}$ $s_t = s_{t-m} + \gamma \varepsilon_t / (\ell_{t-1} + \phi b_{t-1})$

MULTIPLICATIVE ERROR MODELS

Trend	Seasonal		
	N	A	M
N	$y_t = \ell_{t-1}(1 + \varepsilon_t)$ $\ell_t = \ell_{t-1}(1 + \alpha \varepsilon_t)$	$y_t = (\ell_{t-1} + s_{t-m})(1 + \varepsilon_t)$ $\ell_t = \ell_{t-1} + \alpha(\ell_{t-1} + s_{t-m})\varepsilon_t$ $s_t = s_{t-m} + \gamma(\ell_{t-1} + s_{t-m})\varepsilon_t$	$y_t = \ell_{t-1} s_{t-m}(1 + \varepsilon_t)$ $\ell_t = \ell_{t-1}(1 + \alpha \varepsilon_t)$ $s_t = s_{t-m}(1 + \gamma \varepsilon_t)$
A	$y_t = (\ell_{t-1} + b_{t-1})(1 + \varepsilon_t)$ $\ell_t = (\ell_{t-1} + b_{t-1})(1 + \alpha \varepsilon_t)$ $b_t = b_{t-1} + \beta(\ell_{t-1} + b_{t-1})\varepsilon_t$	$y_t = (\ell_{t-1} + b_{t-1} + s_{t-m})(1 + \varepsilon_t)$ $\ell_t = \ell_{t-1} + b_{t-1} + \alpha(\ell_{t-1} + b_{t-1} + s_{t-m})\varepsilon_t$ $b_t = b_{t-1} + \beta(\ell_{t-1} + b_{t-1} + s_{t-m})\varepsilon_t$ $s_t = s_{t-m} + \gamma(\ell_{t-1} + b_{t-1} + s_{t-m})\varepsilon_t$	$y_t = (\ell_{t-1} + b_{t-1})s_{t-m}(1 + \varepsilon_t)$ $\ell_t = (\ell_{t-1} + b_{t-1})(1 + \alpha \varepsilon_t)$ $b_t = b_{t-1} + \beta(\ell_{t-1} + b_{t-1})\varepsilon_t$ $s_t = s_{t-m}(1 + \gamma \varepsilon_t)$
Ad	$y_t = (\ell_{t-1} + \phi b_{t-1})(1 + \varepsilon_t)$ $\ell_t = (\ell_{t-1} + \phi b_{t-1})(1 + \alpha \varepsilon_t)$ $b_t = \phi b_{t-1} + \beta(\ell_{t-1} + \phi b_{t-1})\varepsilon_t$	$y_t = (\ell_{t-1} + \phi b_{t-1} + s_{t-m})(1 + \varepsilon_t)$ $\ell_t = \ell_{t-1} + \phi b_{t-1} + \alpha(\ell_{t-1} + \phi b_{t-1} + s_{t-m})\varepsilon_t$ $b_t = \phi b_{t-1} + \beta(\ell_{t-1} + \phi b_{t-1} + s_{t-m})\varepsilon_t$ $s_t = s_{t-m} + \gamma(\ell_{t-1} + \phi b_{t-1} + s_{t-m})\varepsilon_t$	$y_t = (\ell_{t-1} + \phi b_{t-1})s_{t-m}(1 + \varepsilon_t)$ $\ell_t = (\ell_{t-1} + \phi b_{t-1})(1 + \alpha \varepsilon_t)$ $b_t = \phi b_{t-1} + \beta(\ell_{t-1} + \phi b_{t-1})\varepsilon_t$ $s_t = s_{t-m}(1 + \gamma \varepsilon_t)$

Method or Model?

- A forecasting **method** is a numerical procedure for point forecasts.
- A forecasting **model** describes the stochastic data-generating process and supports prediction intervals, distributional forecasts, likelihood, and AIC.



Obtaining probabilistic forecasts

Implication for inventory

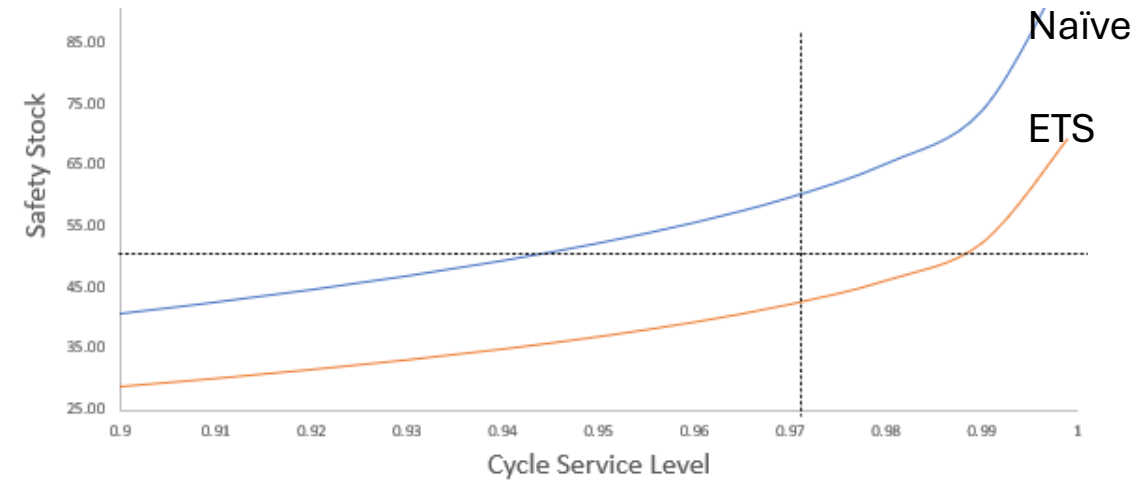
Increase distributional accuracy → less safety stock

```
# implication for inventory
fit <- yTrain %>%
  model(mEts = ETS(Demand),
        mNaive = NAIVE(Demand))

fit %>%
  forecast(h=3) %>%
  accuracy(yFan, list(qs=quantile_score), probs=0.90)
```

.model	.type	qs
<chr>	<chr>	<dbl>
mEts	Test	5.69
mNaive	Test	10.4

.model	Year	Demand
<chr>	<dbl>	<dist>
mEts	2008	N(36, 314)
mNaive	2008	N(30, 867)



Quantile forecast for aggregate demand

Inventory application

- Suppose that we have observed until 2010 Q2 and that the selected models are ETS and NN.
- Stock must cover the last two quarters: The aggregated demand
- **Aggregate demand** = (Demand 2010 Q3) + (Demand 2010 Q4)

$$D^e = D_3 + D_4$$
- Here, D_3 and D_4 are forecast random variables for 2010Q3 and 2010Q4.
- For ETS, the aggregate demand D^e (Total_Demand_Dist) is normally distributed.
- The 97th quantile for the aggregated demand is

```
decision_report
```

A tibble: 2 × 6

.model	Total_Demand_Dist	Mean_Total	Total_Variance	Total_SD	Required_Stock_97
<chr>	<dist>	<dbl>	<dbl>	<dbl>	<dbl>
NN	sample[5000]	907.	204.	14.3	933.
ets	N(891, 292)	891.	292.	17.1	923.

```
# Example: Beer Australian production
recent_production <- aus_production %>%
  filter(year(Quarter) >= 2000)

# Suppose that we have selected two models, ETS and NN
set.seed(1)
beer_fit <- recent_production %>%
  model(
    ets = ETS(Beer),
    NN = NNETAR(Beer)
  )
beer_fc <- beer_fit %>%
  forecast(h=2)

# distributional forecasts
library(distributional)
library(dplyr)

beer_stats <- beer_fc %>%
  as_tibble() %>%
  mutate(
    Mean = mean(Beer),
    Variance = variance(Beer),
    SD = sqrt(Variance),
    P97 = quantile(Beer, 0.97)
  ) %>%
  select(.model, Quarter, Mean, Variance, SD, P97)

beer_stats_formatted <- beer_stats %>%
  mutate(across(where(is.numeric), ~ sprintf("%.2f", .)))

# making decision
decision_report <- beer_fc %>%
  as_tibble() %>%
  group_by(.model) %>%
  summarise(
    Total_Demand_Dist = sum(Beer),
    Mean_Total = sum(.mean)
  ) %>%
  mutate(
    Total_Variance = variance(Total_Demand_Dist),
    Total_SD = sqrt(Total_Variance),
    Required_Stock_97 = quantile(Total_Demand_Dist, 0.97)
  )
```

$$F^{-1}(0.97, 891, \sqrt{292}) \approx 923.$$

$$\text{Excel: } =\text{NORM.INV}(0.97, 891, \sqrt{292})$$

Warning!

Do not add quantiles. Aggregate first, then take the quantile.

```
beer_stats_formatted
```

```
A tibble: 4 x 6
```

.model	Quarter	Mean	Variance	SD	P97
<chr>	<qtr>	<chr>	<chr>	<chr>	<chr>
ets	2010 Q3	406.54	120.08	10.96	427.15
ets	2010 Q4	484.56	171.43	13.09	509.18

Incorrect approach:

1. Find the 0.97 quantile of D_3
2. Find the 0.97 quantile of D_4
3. Add the two quantiles. (This is wrong because quantiles are generally not additive.)

```
decision_report
```

```
A tibble: 2 x 6
```

.model	Total_Demand_Dist	Mean_Total	Total_Variance	Total_SD	Required_Stock_97
<chr>	<dist>	<dbl>	<dbl>	<dbl>	<dbl>
NN	sample[5000]	907.	204.	14.3	933.
ets	N(891, 292)	891.	292.	17.1	923.

- Note that for ETS, the $var(D^e) \neq var(D_3) + var(D_4)$
- This is because

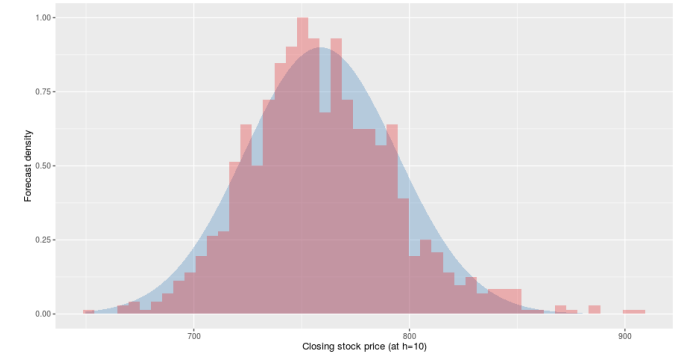
$$var(D^e) = var(D_3 + D_4) = var(D_3) + var(D_4) + 2 cov(D_3, D_4)$$

Correct approach

1. Find the distribution of aggregate demand (Total_Demand_Dist)
2. Find the 0.97 quantile of the aggregate demand distribution.

Distributional forecast (preview)

- Parametric distribution estimation
 - Assume data follows a specific shape, e.g., $N(\mu, \sigma^2)$
 - The model predicts the parameters of that shape.
Point forecast $\rightarrow \mu$ Forecast standard deviation $\rightarrow \sigma$
 - Classical statistical models, e.g., simple methods, ETS, ARIMA assume $e_t \sim N(0, \sigma^2)$.
- Quantile regression
 - Predict a set of specific quantiles, e.g., 0.01, 0.02, ..., 0.98, 0.99
 - Train the model using the pinball loss function
 - Quantile regression, quantile random forest
- Monte Carlo simulation
 - Bootstrapping: uncorrelated residuals with constant variance (homoskedasticity).
 - $y_{T+1}^* = y_T + e_{T+1}^*$ where e_{T+1}^* is a randomly sampled error from the past.
- ϕ -Divergence concept
 - Measure the “distance” between two probability distributions.
 - Distributional Robust Optimization



Quantile score: Estimating smoothing parameter α^* ETS(A,N,N)

```
#####
# MLE & Quantile score
#####
y <- tsibble(
  Quarter = yearquarter("2010 Q3") + 0:13,
  obs = c(1212, 1321, 1278, 1341, 1257, 1341, 1257, 1287,
          1189, 1111, 1145, 1150, 1298, 1331),
  index = Quarter
)

fit <- y %>%
  filter(year(Quarter) <= 2012) %>%
  model(myETS=ETS(Obs ~ error("A") + trend("N") + season("N")))
report(fit)

fc <- fit %>%
  forecast(h=4)
accuracy(fc, y, list(qs=quantile_score), probs=0.9)

> fc
# A tibble: 4 x 4 [1Q]
# Key:   .model [1]
  .model Quarter      obs .mean
  <chr>   <qtr>      <dbl> <dbl>
1 myETS  2013 Q1  N(1154, 5926) 1154.
2 myETS  2013 Q2  N(1154, 8205) 1154.
3 myETS  2013 Q3  N(1154, 10483) 1154.
4 myETS  2013 Q4  N(1154, 12762) 1154.
> accuracy(fc, y, list(qs=quantile_score), probs=0.9)
# A tibble: 1 x 3
  .model .type    qs
  <chr>   <chr> <dbl>
1 myETS  Test   31.3
```

Note that the point FCs are identical for all h , but the quantile forecast increases as h increases

ETS(A,N,N)				k
Initial Level		1249.2866258		
Smoothing parameter		0.6201022		
t	Observation	Level	Fitted	
0		1249.29		
1	1212	1226.17	1249.29	
2	1321	1284.97	1226.17	
3	1278	1280.65	1284.97	
4	1341	1318.07	1280.65	
5	1257	1280.20	1318.07	
6	1341	1317.90	1280.20	
7	1257	1280.14	1317.90	
8	1287	1284.39	1280.14	
9	1189	1225.24	1284.39	
10	1111	1154.40	1225.24	
11				
12				
13				
14				
T		10		
Residual sum of of squares (RSS) or SEE				47407.85
σ^2				5925.981008
σ				76.98039366

Sheet ETS_ANN

	Quantile	0.90
	Actual	Quantile FC
h=1	1145	1253.1
h=2	1150	1270.5
h=3	1298	1285.6
h=4	1331	1299.2
Quantile Score		31.3

	24.1
0.1	24.129
0.2	21.725
0.3	19.408
0.4	22.488
0.5	25.559
0.6	30.716
0.7	32.225
0.8	29.800
0.9	23.005

The smoothing parameter that minimizes the quantile score (at $p = 0.90$) is $\alpha^* = 0.3$
Homework (Optional): Choose α to minimize the quantile score at $p = 0.9547$

Hw

Section 8.8 Problem 1a

- Consider the number of pigs slaughtered in Victoria, available in the `aus_livestock` dataset.
- Use the `ETS()` function in R to estimate the equivalent model for **simple** exponential smoothing.
- From the report, find the optimal values of α and ℓ_0 (determined by R), and generate forecasts for the next four months.

```
pigs <- aus_livestock %>%  
  filter(Animal == "Pigs", State == "Victoria")  
  
fit <- pigs %>%  
  model(****)  
report(fit)  
  
fit %>%  
  forecast(****)
```

Time	Forecast
Jan 2019	
Feb 2019	
Mar 2019	
Apr 2019	

Appendix

Theory: State-Space Models

Innovations state space models for exponential smoothing

- **Innovations state space** model consists of
 - 1) **Measurement equation** that describes the observed data, i.e., the relationship between observation & states.
 - 2) **State/transition equations** that describe how the *unobserved* components or states (level, trend, season) change over time.
- “innovations” comes from the fact that all equations use the same random error process $\{\epsilon_t\}$

ADDITIVE ERROR MODELS

Trend	N	Seasonal A	M
N	$y_t = \ell_{t-1} + \epsilon_t$ $\ell_t = \ell_{t-1} + \alpha \epsilon_t$	$y_t = \ell_{t-1} + s_{t-m} + \epsilon_t$ $\ell_t = \ell_{t-1} + \alpha \epsilon_t$ $s_t = s_{t-m} + \gamma \epsilon_t$	$y_t = \ell_{t-1} s_{t-m} + \epsilon_t$ $\ell_t = \ell_{t-1} + \alpha \epsilon_t / s_{t-m}$ $s_t = s_{t-m} + \gamma \epsilon_t / \ell_{t-1}$
A	$y_t = \ell_{t-1} + b_{t-1} + \epsilon_t$ $\ell_t = \ell_{t-1} + b_{t-1} + \alpha \epsilon_t$ $b_t = b_{t-1} + \beta \epsilon_t$	$y_t = \ell_{t-1} + b_{t-1} + s_{t-m} + \epsilon_t$ $\ell_t = \ell_{t-1} + b_{t-1} + \alpha \epsilon_t$ $b_t = b_{t-1} + \beta \epsilon_t$ $s_t = s_{t-m} + \gamma \epsilon_t$	$y_t = (\ell_{t-1} + b_{t-1}) s_{t-m} + \epsilon_t$ $\ell_t = \ell_{t-1} + b_{t-1} + \alpha \epsilon_t / s_{t-m}$ $b_t = b_{t-1} + \beta \epsilon_t / s_{t-m}$ $s_t = s_{t-m} + \gamma \epsilon_t / (\ell_{t-1} + b_{t-1})$
A _d	$y_t = \ell_{t-1} + \phi b_{t-1} + \epsilon_t$ $\ell_t = \ell_{t-1} + \phi b_{t-1} + \alpha \epsilon_t$ $b_t = \phi b_{t-1} + \beta \epsilon_t$	$y_t = \ell_{t-1} + \phi b_{t-1} + s_{t-m} + \epsilon_t$ $\ell_t = \ell_{t-1} + \phi b_{t-1} + \alpha \epsilon_t$ $b_t = \phi b_{t-1} + \beta \epsilon_t$ $s_t = s_{t-m} + \gamma \epsilon_t$	$y_t = (\ell_{t-1} + \phi b_{t-1}) s_{t-m} + \epsilon_t$ $\ell_t = \ell_{t-1} + \phi b_{t-1} + \alpha \epsilon_t / s_{t-m}$ $b_t = \phi b_{t-1} + \beta \epsilon_t / s_{t-m}$ $s_t = s_{t-m} + \gamma \epsilon_t / (\ell_{t-1} + \phi b_{t-1})$

State space equations for each of the models in the ETS framework

ETS(·,·,·) for (Error, Trend, Seasonal)

ADDITIVE ERROR MODELS

Trend	Seasonal		
	N	A	M
N	$y_t = \ell_{t-1} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \alpha \varepsilon_t$	$y_t = \ell_{t-1} + s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \alpha \varepsilon_t$ $s_t = s_{t-m} + \gamma \varepsilon_t$	$y_t = \ell_{t-1} s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \alpha \varepsilon_t / s_{t-m}$ $s_t = s_{t-m} + \gamma \varepsilon_t / \ell_{t-1}$
A	$y_t = \ell_{t-1} + b_{t-1} + \varepsilon_t$ $\ell_t = \ell_{t-1} + b_{t-1} + \alpha \varepsilon_t$ $b_t = b_{t-1} + \beta \varepsilon_t$	$y_t = \ell_{t-1} + b_{t-1} + s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + b_{t-1} + \alpha \varepsilon_t$ $b_t = b_{t-1} + \beta \varepsilon_t$ $s_t = s_{t-m} + \gamma \varepsilon_t$	$y_t = (\ell_{t-1} + b_{t-1}) s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + b_{t-1} + \alpha \varepsilon_t / s_{t-m}$ $b_t = b_{t-1} + \beta \varepsilon_t / s_{t-m}$ $s_t = s_{t-m} + \gamma \varepsilon_t / (\ell_{t-1} + b_{t-1})$
A_d	$y_t = \ell_{t-1} + \phi b_{t-1} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \phi b_{t-1} + \alpha \varepsilon_t$ $b_t = \phi b_{t-1} + \beta \varepsilon_t$	$y_t = \ell_{t-1} + \phi b_{t-1} + s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \phi b_{t-1} + \alpha \varepsilon_t$ $b_t = \phi b_{t-1} + \beta \varepsilon_t$ $s_t = s_{t-m} + \gamma \varepsilon_t$	$y_t = (\ell_{t-1} + \phi b_{t-1}) s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \phi b_{t-1} + \alpha \varepsilon_t / s_{t-m}$ $b_t = \phi b_{t-1} + \beta \varepsilon_t / s_{t-m}$ $s_t = s_{t-m} + \gamma \varepsilon_t / (\ell_{t-1} + \phi b_{t-1})$

ANN

MNN

MULTIPLICATIVE ERROR MODELS

Trend	Seasonal		
	N	A	M
N	$y_t = \ell_{t-1} (1 + \varepsilon_t)$ $\ell_t = \ell_{t-1} (1 + \alpha \varepsilon_t)$	$y_t = (\ell_{t-1} + s_{t-m}) (1 + \varepsilon_t)$ $\ell_t = \ell_{t-1} + \alpha (\ell_{t-1} + s_{t-m}) \varepsilon_t$ $s_t = s_{t-m} + \gamma (\ell_{t-1} + s_{t-m}) \varepsilon_t$	$y_t = \ell_{t-1} s_{t-m} (1 + \varepsilon_t)$ $\ell_t = \ell_{t-1} (1 + \alpha \varepsilon_t)$ $s_t = s_{t-m} (1 + \gamma \varepsilon_t)$
A	$y_t = (\ell_{t-1} + b_{t-1}) (1 + \varepsilon_t)$ $\ell_t = (\ell_{t-1} + b_{t-1}) (1 + \alpha \varepsilon_t)$ $b_t = b_{t-1} + \beta (\ell_{t-1} + b_{t-1}) \varepsilon_t$	$y_t = (\ell_{t-1} + b_{t-1} + s_{t-m}) (1 + \varepsilon_t)$ $\ell_t = \ell_{t-1} + b_{t-1} + \alpha (\ell_{t-1} + b_{t-1} + s_{t-m}) \varepsilon_t$ $b_t = b_{t-1} + \beta (\ell_{t-1} + b_{t-1} + s_{t-m}) \varepsilon_t$ $s_t = s_{t-m} + \gamma (\ell_{t-1} + b_{t-1} + s_{t-m}) \varepsilon_t$	$y_t = (\ell_{t-1} + b_{t-1}) s_{t-m} (1 + \varepsilon_t)$ $\ell_t = (\ell_{t-1} + b_{t-1}) (1 + \alpha \varepsilon_t)$ $b_t = b_{t-1} + \beta (\ell_{t-1} + b_{t-1}) \varepsilon_t$ $s_t = s_{t-m} (1 + \gamma \varepsilon_t)$
A_d	$y_t = (\ell_{t-1} + \phi b_{t-1}) (1 + \varepsilon_t)$ $\ell_t = (\ell_{t-1} + \phi b_{t-1}) (1 + \alpha \varepsilon_t)$ $b_t = \phi b_{t-1} + \beta (\ell_{t-1} + \phi b_{t-1}) \varepsilon_t$	$y_t = (\ell_{t-1} + \phi b_{t-1} + s_{t-m}) (1 + \varepsilon_t)$ $\ell_t = \ell_{t-1} + \phi b_{t-1} + \alpha (\ell_{t-1} + \phi b_{t-1} + s_{t-m}) \varepsilon_t$ $b_t = \phi b_{t-1} + \beta (\ell_{t-1} + \phi b_{t-1} + s_{t-m}) \varepsilon_t$ $s_t = s_{t-m} + \gamma (\ell_{t-1} + \phi b_{t-1} + s_{t-m}) \varepsilon_t$	$y_t = (\ell_{t-1} + \phi b_{t-1}) s_{t-m} (1 + \varepsilon_t)$ $\ell_t = (\ell_{t-1} + \phi b_{t-1}) (1 + \alpha \varepsilon_t)$ $b_t = \phi b_{t-1} + \beta (\ell_{t-1} + \phi b_{t-1}) \varepsilon_t$ $s_t = s_{t-m} (1 + \gamma \varepsilon_t)$

AAN

MAA

AAA

ETS($\cdot, \cdot, \cdot$) for (Error, Trend, Seasonal)

ADDITIVE ERROR MODELS

Trend	N	Seasonal A	M
N	$y_t = \ell_{t-1} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \alpha \varepsilon_t$	$y_t = \ell_{t-1} + s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \alpha \varepsilon_t$ $s_t = s_{t-m} + \gamma \varepsilon_t$	$y_t = \ell_{t-1} s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \alpha \varepsilon_t / s_{t-m}$ $s_t = s_{t-m} + \gamma \varepsilon_t / \ell_{t-1}$
A	$y_t = \ell_{t-1} + b_{t-1} + \varepsilon_t$ $\ell_t = \ell_{t-1} + b_{t-1} + \alpha \varepsilon_t$ $b_t = b_{t-1} + \beta \varepsilon_t$	$y_t = \ell_{t-1} + b_{t-1} + s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + b_{t-1} + \alpha \varepsilon_t$ $b_t = b_{t-1} + \beta \varepsilon_t$ $s_t = s_{t-m} + \gamma \varepsilon_t$	$y_t = \ell_{t-1} s_{t-m} + b_{t-1} + \varepsilon_t$ $\ell_t = \ell_{t-1} + b_{t-1} + \alpha \varepsilon_t / s_{t-m}$ $b_t = b_{t-1} + \beta \varepsilon_t / s_{t-m}$ $s_t = s_{t-m} + \gamma \varepsilon_t / (\ell_{t-1} + b_{t-1})$
Ad	$y_t = \ell_{t-1} + \phi b_{t-1} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \phi b_{t-1} + \alpha \varepsilon_t$ $b_t = \phi b_{t-1} + \beta \varepsilon_t$	$y_t = \ell_{t-1} + \phi b_{t-1} + s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \phi b_{t-1} + \alpha \varepsilon_t$ $b_t = \phi b_{t-1} + \beta \varepsilon_t$ $s_t = s_{t-m} + \gamma \varepsilon_t$	$y_t = (\ell_{t-1} + \phi b_{t-1}) s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \phi b_{t-1} + \alpha \varepsilon_t / s_{t-m}$ $b_t = \phi b_{t-1} + \beta \varepsilon_t / s_{t-m}$ $s_t = s_{t-m} + \gamma \varepsilon_t / (\ell_{t-1} + \phi b_{t-1})$

MULTIPLICATIVE ERROR MODELS

Trend	N	Seasonal A	M
N	$y_t = \ell_{t-1} (1 + \varepsilon_t)$ $\ell_t = \ell_{t-1} (1 + \alpha \varepsilon_t)$	$y_t = (\ell_{t-1} + s_{t-m}) (1 + \varepsilon_t)$ $\ell_t = \ell_{t-1} + \alpha (\ell_{t-1} + s_{t-m}) \varepsilon_t$ $s_t = s_{t-m} + \gamma (\ell_{t-1} + s_{t-m}) \varepsilon_t$	$y_t = \ell_{t-1} s_{t-m} (1 + \varepsilon_t)$ $\ell_t = \ell_{t-1} (1 + \alpha \varepsilon_t)$ $s_t = s_{t-m} (1 + \gamma \varepsilon_t)$
A	$y_t = (\ell_{t-1} + b_{t-1}) (1 + \varepsilon_t)$ $\ell_t = (\ell_{t-1} + b_{t-1}) (1 + \alpha \varepsilon_t)$ $b_t = b_{t-1} + \beta (\ell_{t-1} + b_{t-1}) \varepsilon_t$	$y_t = (\ell_{t-1} + b_{t-1} + s_{t-m}) (1 + \varepsilon_t)$ $\ell_t = \ell_{t-1} + b_{t-1} + \alpha (\ell_{t-1} + b_{t-1} + s_{t-m}) \varepsilon_t$ $b_t = b_{t-1} + \beta (\ell_{t-1} + b_{t-1} + s_{t-m}) \varepsilon_t$ $s_t = s_{t-m} + \gamma (\ell_{t-1} + b_{t-1} + s_{t-m}) \varepsilon_t$	$y_t = (\ell_{t-1} + b_{t-1}) s_{t-m} (1 + \varepsilon_t)$ $\ell_t = (\ell_{t-1} + b_{t-1}) (1 + \alpha \varepsilon_t)$ $b_t = b_{t-1} + \beta (\ell_{t-1} + b_{t-1}) \varepsilon_t$ $s_t = s_{t-m} (1 + \gamma \varepsilon_t)$
Ad	$y_t = (\ell_{t-1} + \phi b_{t-1}) (1 + \varepsilon_t)$ $\ell_t = (\ell_{t-1} + \phi b_{t-1}) (1 + \alpha \varepsilon_t)$ $b_t = \phi b_{t-1} + \beta (\ell_{t-1} + \phi b_{t-1}) \varepsilon_t$	$y_t = (\ell_{t-1} + \phi b_{t-1} + s_{t-m}) (1 + \varepsilon_t)$ $\ell_t = \ell_{t-1} + \phi b_{t-1} + \alpha (\ell_{t-1} + \phi b_{t-1} + s_{t-m}) \varepsilon_t$ $b_t = \phi b_{t-1} + \beta (\ell_{t-1} + \phi b_{t-1} + s_{t-m}) \varepsilon_t$ $s_t = s_{t-m} + \gamma (\ell_{t-1} + \phi b_{t-1} + s_{t-m}) \varepsilon_t$	$y_t = (\ell_{t-1} + \phi b_{t-1}) s_{t-m} (1 + \varepsilon_t)$ $\ell_t = (\ell_{t-1} + \phi b_{t-1}) (1 + \alpha \varepsilon_t)$ $b_t = \phi b_{t-1} + \beta (\ell_{t-1} + \phi b_{t-1}) \varepsilon_t$ $s_t = s_{t-m} (1 + \gamma \varepsilon_t)$

3 combinations of (Error, Trend, Seasonal), namely

ETS(A,N,M), ETS(A,A,M), ETS(A,Ad,M)

can lead to numerical difficulties due to division by values potentially close to zero in the state equations.

We do not consider these combinations.

ETS($\cdot, \cdot, \cdot$) for (Error, Trend, Seasonal)

ADDITIVE ERROR MODELS

Trend	N	Seasonal A	M
N	$y_t = \ell_{t-1} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \alpha \varepsilon_t$	$y_t = \ell_{t-1} + s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \alpha \varepsilon_t$ $s_t = s_{t-m} + \gamma \varepsilon_t$	$y_t = \ell_{t-1} s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \alpha \varepsilon_t / s_{t-m}$ $s_t = s_{t-m} + \gamma \varepsilon_t / \ell_{t-1}$
A	$y_t = \ell_{t-1} + b_{t-1} + \varepsilon_t$ $\ell_t = \ell_{t-1} + b_{t-1} + \alpha \varepsilon_t$ $b_t = b_{t-1} + \beta \varepsilon_t$	$y_t = \ell_{t-1} + b_{t-1} + s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + b_{t-1} + \alpha \varepsilon_t$ $b_t = b_{t-1} + \beta \varepsilon_t$ $s_t = s_{t-m} + \gamma \varepsilon_t$	$y_t = \ell_{t-1} s_{t-m} + b_{t-1} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \alpha \varepsilon_t / s_{t-m}$ $b_t = b_{t-1} + \beta \varepsilon_t / s_{t-m}$ $s_t = s_{t-m} + \gamma \varepsilon_t / (\ell_{t-1} + b_{t-1})$
Λ_d	$y_t = \ell_{t-1} + \phi b_{t-1} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \phi b_{t-1} + \alpha \varepsilon_t$ $b_t = \phi b_{t-1} + \beta \varepsilon_t$	$y_t = \ell_{t-1} + \phi b_{t-1} + s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \phi b_{t-1} + \alpha \varepsilon_t$ $b_t = \phi b_{t-1} + \beta \varepsilon_t$ $s_t = s_{t-m} + \gamma \varepsilon_t$	$y_t = (\ell_{t-1} + \phi b_{t-1}) s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \phi b_{t-1} + \alpha \varepsilon_t / s_{t-m}$ $b_t = \phi b_{t-1} + \beta \varepsilon_t / s_{t-m}$ $s_t = s_{t-m} + \gamma \varepsilon_t / (\ell_{t-1} + \phi b_{t-1})$

MULTIPLICATIVE ERROR MODELS

Trend	N	Seasonal A	M
N	$y_t = \ell_{t-1} (1 + \varepsilon_t)$ $\ell_t = \ell_{t-1} (1 + \alpha \varepsilon_t)$	$y_t = (\ell_{t-1} + s_{t-m}) (1 + \varepsilon_t)$ $\ell_t = \ell_{t-1} + \alpha (\ell_{t-1} + s_{t-m}) \varepsilon_t$ $s_t = s_{t-m} + \gamma (\ell_{t-1} + s_{t-m}) \varepsilon_t$	$y_t = \ell_{t-1} s_{t-m} (1 + \varepsilon_t)$ $\ell_t = \ell_{t-1} (1 + \alpha \varepsilon_t)$ $s_t = s_{t-m} (1 + \gamma \varepsilon_t)$
A	$y_t = (\ell_{t-1} + b_{t-1}) (1 + \varepsilon_t)$ $\ell_t = (\ell_{t-1} + b_{t-1}) (1 + \alpha \varepsilon_t)$ $b_t = b_{t-1} + \beta (\ell_{t-1} + b_{t-1}) \varepsilon_t$	$y_t = (\ell_{t-1} + b_{t-1} + s_{t-m}) (1 + \varepsilon_t)$ $\ell_t = \ell_{t-1} + b_{t-1} + \alpha (\ell_{t-1} + b_{t-1} + s_{t-m}) \varepsilon_t$ $b_t = b_{t-1} + \beta (\ell_{t-1} + b_{t-1} + s_{t-m}) \varepsilon_t$ $s_t = s_{t-m} + \gamma (\ell_{t-1} + b_{t-1} + s_{t-m}) \varepsilon_t$	$y_t = (\ell_{t-1} + b_{t-1}) s_{t-m} (1 + \varepsilon_t)$ $\ell_t = (\ell_{t-1} + b_{t-1}) (1 + \alpha \varepsilon_t)$ $b_t = b_{t-1} + \beta (\ell_{t-1} + b_{t-1}) \varepsilon_t$ $s_t = s_{t-m} (1 + \gamma \varepsilon_t)$
Λ_d	$y_t = (\ell_{t-1} + \phi b_{t-1}) (1 + \varepsilon_t)$ $\ell_t = (\ell_{t-1} + \phi b_{t-1}) (1 + \alpha \varepsilon_t)$ $b_t = \phi b_{t-1} + \beta (\ell_{t-1} + \phi b_{t-1}) \varepsilon_t$	$y_t = (\ell_{t-1} + \phi b_{t-1} + s_{t-m}) (1 + \varepsilon_t)$ $\ell_t = \ell_{t-1} + \phi b_{t-1} + \alpha (\ell_{t-1} + \phi b_{t-1} + s_{t-m}) \varepsilon_t$ $b_t = \phi b_{t-1} + \beta (\ell_{t-1} + \phi b_{t-1} + s_{t-m}) \varepsilon_t$ $s_t = s_{t-m} + \gamma (\ell_{t-1} + \phi b_{t-1} + s_{t-m}) \varepsilon_t$	$y_t = (\ell_{t-1} + \phi b_{t-1}) s_{t-m} (1 + \varepsilon_t)$ $\ell_t = (\ell_{t-1} + \phi b_{t-1}) (1 + \alpha \varepsilon_t)$ $b_t = \phi b_{t-1} + \beta (\ell_{t-1} + \phi b_{t-1}) \varepsilon_t$ $s_t = s_{t-m} (1 + \gamma \varepsilon_t)$

Models with multiplicative errors are useful when data are *strictly positive*.

Models with multiplicative errors are not numerically stable when data contain zeros or negative values.

ETS($\cdot, \cdot, \cdot$) for (Error, Trend, Seasonal)

ADDITIVE ERROR MODELS			
Trend	Seasonal		
	N	A	M
N	$y_t = \ell_{t-1} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \alpha \varepsilon_t$	$y_t = \ell_{t-1} + s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \alpha \varepsilon_t$ $s_t = s_{t-m} + \gamma \varepsilon_t$	$y_t = \ell_{t-1} s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \alpha \varepsilon_t / s_{t-m}$ $s_t = s_{t-m} + \gamma \varepsilon_t / \ell_{t-1}$
A	$y_t = \ell_{t-1} + b_{t-1} + \varepsilon_t$ $\ell_t = \ell_{t-1} + b_{t-1} + \alpha \varepsilon_t$ $b_t = b_{t-1} + \beta \varepsilon_t$	$y_t = \ell_{t-1} + b_{t-1} + s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + b_{t-1} + \alpha \varepsilon_t$ $b_t = b_{t-1} + \beta \varepsilon_t$ $s_t = s_{t-m} + \gamma \varepsilon_t$	$y_t = \ell_{t-1} b_{t-1} s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \alpha \varepsilon_t / (b_{t-1} s_{t-m})$ $b_t = b_{t-1} + \beta \varepsilon_t / s_{t-m}$ $s_t = s_{t-m} + \gamma \varepsilon_t / (\ell_{t-1} b_{t-1})$
Λ_d	$y_t = \ell_{t-1} + \phi b_{t-1} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \phi b_{t-1} + \alpha \varepsilon_t$ $b_t = \phi b_{t-1} + \beta \varepsilon_t$	$y_t = \ell_{t-1} + \phi b_{t-1} + s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \phi b_{t-1} + \alpha \varepsilon_t$ $b_t = \phi b_{t-1} + \beta \varepsilon_t$ $s_t = s_{t-m} + \gamma \varepsilon_t$	$y_t = (\ell_{t-1} + \phi b_{t-1}) s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \phi b_{t-1} + \alpha \varepsilon_t / s_{t-m}$ $b_t = \phi b_{t-1} + \beta \varepsilon_t / s_{t-m}$ $s_t = s_{t-m} + \gamma \varepsilon_t / (\ell_{t-1} + \phi b_{t-1})$

Suppose that time series is not strictly positive.

Multiplicative error models will not be considered.

Only the six fully additive models will be applied.

MULTIPLICATIVE ERROR MODELS			
Trend	Seasonal		
	N	A	M
N	$y_t = \ell_{t-1} (1 + \varepsilon_t)$ $\ell_t = \ell_{t-1} (1 + \alpha \varepsilon_t)$	$y_t = (\ell_{t-1} + s_{t-m}) (1 + \varepsilon_t)$ $\ell_t = \ell_{t-1} + \alpha (\ell_{t-1} + s_{t-m}) \varepsilon_t$ $s_t = s_{t-m} + \gamma (\ell_{t-1} + s_{t-m}) \varepsilon_t$	$y_t = \ell_{t-1} s_{t-m} (1 + \varepsilon_t)$ $\ell_t = \ell_{t-1} (1 + \alpha \varepsilon_t)$ $s_t = s_{t-m} (1 + \gamma \varepsilon_t)$
A	$y_t = (\ell_{t-1} + b_{t-1}) (1 + \varepsilon_t)$ $\ell_t = (\ell_{t-1} + b_{t-1}) (1 + \alpha \varepsilon_t)$ $b_t = b_{t-1} + \beta (\ell_{t-1} + b_{t-1}) \varepsilon_t$	$y_t = (\ell_{t-1} + b_{t-1} + s_{t-m}) (1 + \varepsilon_t)$ $\ell_t = \ell_{t-1} + b_{t-1} + \alpha (\ell_{t-1} + b_{t-1} + s_{t-m}) \varepsilon_t$ $b_t = b_{t-1} + \beta (\ell_{t-1} + b_{t-1} + s_{t-m}) \varepsilon_t$ $s_t = s_{t-m} + \gamma (\ell_{t-1} + b_{t-1} + s_{t-m}) \varepsilon_t$	$y_t = (\ell_{t-1} + b_{t-1}) s_{t-m} (1 + \varepsilon_t)$ $\ell_t = (\ell_{t-1} + b_{t-1}) (1 + \alpha \varepsilon_t)$ $b_t = b_{t-1} + \beta (\ell_{t-1} + b_{t-1}) \varepsilon_t$ $s_t = s_{t-m} (1 + \gamma \varepsilon_t)$
Λ_d	$y_t = (\ell_{t-1} + \phi b_{t-1}) (1 + \varepsilon_t)$ $\ell_t = (\ell_{t-1} + \phi b_{t-1}) (1 + \alpha \varepsilon_t)$ $b_t = \phi b_{t-1} + \beta (\ell_{t-1} + \phi b_{t-1}) \varepsilon_t$	$y_t = (\ell_{t-1} + \phi b_{t-1} + s_{t-m}) (1 + \varepsilon_t)$ $\ell_t = \ell_{t-1} + \phi b_{t-1} + \alpha (\ell_{t-1} + \phi b_{t-1} + s_{t-m}) \varepsilon_t$ $b_t = \phi b_{t-1} + \beta (\ell_{t-1} + \phi b_{t-1} + s_{t-m}) \varepsilon_t$ $s_t = s_{t-m} + \gamma (\ell_{t-1} + \phi b_{t-1} + s_{t-m}) \varepsilon_t$	$y_t = (\ell_{t-1} + \phi b_{t-1}) s_{t-m} (1 + \varepsilon_t)$ $\ell_t = (\ell_{t-1} + \phi b_{t-1}) (1 + \alpha \varepsilon_t)$ $b_t = \phi b_{t-1} + \beta (\ell_{t-1} + \phi b_{t-1}) \varepsilon_t$ $s_t = s_{t-m} (1 + \gamma \varepsilon_t)$

State space model*

ADDITIVE ERROR MODELS

Trend	Seasonal		
	N	A	M
N	$y_t = \ell_{t-1} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \alpha \varepsilon_t$	$y_t = \ell_{t-1} + s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \alpha \varepsilon_t$ $s_t = s_{t-m} + \gamma \varepsilon_t$	$y_t = \ell_{t-1} s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \alpha \varepsilon_t / s_{t-m}$ $s_t = s_{t-m} + \gamma \varepsilon_t / \ell_{t-1}$
A	$y_t = \ell_{t-1} + b_{t-1} + \varepsilon_t$ $\ell_t = \ell_{t-1} + b_{t-1} + \alpha \varepsilon_t$ $b_t = b_{t-1} + \beta \varepsilon_t$	$y_t = \ell_{t-1} + b_{t-1} + s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + b_{t-1} + \alpha \varepsilon_t$ $b_t = b_{t-1} + \beta \varepsilon_t$ $s_t = s_{t-m} + \gamma \varepsilon_t$	$y_t = (\ell_{t-1} + b_{t-1}) s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + b_{t-1} + \alpha \varepsilon_t / s_{t-m}$ $b_t = b_{t-1} + \beta \varepsilon_t / s_{t-m}$ $s_t = s_{t-m} + \gamma \varepsilon_t / (\ell_{t-1} + b_{t-1})$
A _d	$y_t = \ell_{t-1} + \phi b_{t-1} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \phi b_{t-1} + \alpha \varepsilon_t$ $b_t = \phi b_{t-1} + \beta \varepsilon_t$	$y_t = \ell_{t-1} + \phi b_{t-1} + s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \phi b_{t-1} + \alpha \varepsilon_t$ $b_t = \phi b_{t-1} + \beta \varepsilon_t$ $s_t = s_{t-m} + \gamma \varepsilon_t$	$y_t = (\ell_{t-1} + \phi b_{t-1}) s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \phi b_{t-1} + \alpha \varepsilon_t / s_{t-m}$ $b_t = \phi b_{t-1} + \beta \varepsilon_t / s_{t-m}$ $s_t = s_{t-m} + \gamma \varepsilon_t / (\ell_{t-1} + \phi b_{t-1})$

ETS(A,A,A)

Measurement equation

$$y_t = \ell_{t-1} + b_{t-1} + s_{t-m} + \varepsilon_t$$

State equations

$$\ell_t = \ell_{t-1} + b_{t-1} + \alpha \varepsilon_t$$

$$b_t = b_{t-1} + \beta \varepsilon_t$$

$$s_t = s_{t-m} + \gamma \varepsilon_t$$

ETS(A,N,N)

Measurement equation

$$y_t =$$

State equations

SES: Error correction form*

Last lecture

- Weighted average form

$$\hat{y}_{T+h|T} = \alpha y_T + (1 - \alpha) \hat{y}_{T|T-1}$$

- Component form

- Forecast eq $\hat{y}_{t+h|t} = \ell_t$
- Smoothing eq $\ell_t = \alpha y_t + (1 - \alpha) \ell_{t-1}$

Today's lecture

- “Error correction” form

$$\ell_t = \ell_{t-1} + \alpha e_t$$

where $e_t = y_t - \hat{y}_{t|t-1}$ is the residual at time t

Initial Level		1249.287		
Smoothing parameter		0.6201		
t	Demand	Level	Fitted	Residual
0		1249.29		
1	1212	1226.17	1249.29	-37.29
2	1321	1284.97	1226.17	94.83
3	1278	1280.65	1284.97	-6.97
4	1341	1318.07	1280.65	60.35
5	1257	1280.20	1318.07	-61.07
6	1341	1317.90	1280.20	60.80
7	1257	1280.14	1317.90	-60.90
8	1287	1284.39	1280.14	6.86
9	1189	1225.24	1284.39	-95.39
10	1111	1154.40	1225.24	-114.24

ETS(A,N,N): SES with additive errors*

- For models with additive errors, assume $e_t = \epsilon_t \sim NID(0, \sigma^2)$

- Measurement (observation) eq: $y_t = \ell_{t-1} + \epsilon_t$

This follows from $e_t = y_t - \hat{y}_{t|t-1} = y_t - \ell_{t-1}$

- State (transition) eq: $\ell_t = \ell_{t-1} + \alpha \epsilon_t$

This follows from the error correction form.

parameter estimation* & model selection

Estimating ETS models

- Alternative to estimating the parameters by minimizing the sum of squared errors (SSE) is to maximize the “**likelihood**”
- The likelihood is the probability of the data arising from the specified model.
- Large likelihood is associated with a good model
- For additive error model, maximizing likelihood gives the same results as minimizing SSE.
 - Minimizing the AIC assuming Gaussian residuals is asymptotically equivalent to minimizing one-step time series cross validation MSE.
- Different results will be obtained for multiplicative models.

Maximum Likelihood Estimator (MLE)*

Sheet ETS_ANN

ETS(A,N,N)



Initial Level 1249.2866258

Smoothing parameter 0.6201022

k 2

t	Observation	Level	Fitted	Residual	Likelihood	Log likelihood	α	SSE	Likelihood
0		1249.29					0.10	47,407.85	2.559E-25
1	1212	1226.17	1249.29	-37.29	0.004609	-5.379794131	0.20	51,543.23	1.685E-25
2	1321	1284.97	1226.17	94.83	0.002426	-6.021322043	0.30	52,228.15	1.577E-25
3	1278	1280.65	1284.97	-6.97	0.005161	-5.266591132	0.40	51,154.44	1.750E-25
4	1341	1318.07	1280.65	60.35	0.003811	-5.569802624	0.50	49,493.33	2.064E-25
5	1257	1280.20	1318.07	-61.07	0.003783	-5.577194715	0.60	48,093.10	2.382E-25
6	1341	1317.90	1280.20	60.80	0.003794	-5.574376128	0.70	47,753.21	2.468E-25
7	1257	1280.14	1317.90	-60.90	0.00379	-5.57544541	0.80	49,255.36	2.114E-25
8	1287	1284.39	1280.14	6.86	0.005162	-5.266463596	0.90	52,165.49	1.587E-25
9	1189	1225.24	1284.39	-95.39	0.002405	-6.03027489			
10	1111	1154.40	1225.24	-114.24	0.001723	-6.363628277			

T 10

Residual sum of of squares (RSS), SSE,

$\sum_{t=1}^T e_t^2$ 47407.85
 $\hat{\sigma}^2$ 5925.981

σ 76.98039

Likelihood 2.56E-25

log(likelihood) -56.6249 -56.6249

AIC = -2*log(likelihood) + 2*k/T 113.6498

```
y <- tsibble(
  Quarter = yearquarter("2010 Q3") + 0:13,
  Obs = c(1212, 1321, 1278, 1341, 1257, 1341,
1257, 1287, 1189, 1111, 1145, 1150, 1298, 1331),
  index = Quarter
)

yTrain <- y %>% filter(year(Quarter) <= 2012)

fit <- yTrain %>%
  model(ets = ETS(Obs))

report(fit)
```

Series: Obs
Model: ETS(A,N,N)
Smoothing parameters:
alpha = 0.6201022

Initial states:
1[0]
1249.287

sigma^2: 5925.981

AIC	AICc	BIC
113.6654	117.6654	114.5732

- The density function of normal distribution with mean μ and standard deviation σ is

$$f(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

- Likelihood $L = \prod_{i=1}^n f(x_i; \mu, \sigma)$
- Here, error is independent and normally distributed with mean 0 and standard deviation σ
- $\hat{\sigma} = \sqrt{\frac{1}{T-k} \sum_{t=1}^T e_t^2}$ where T is the number of fitted values and k is the total number of parameters and initial states.
- For additive error model, maximizing likelihood gives the same results as minimizing SSE.

Information criteria

- For ETS models, Akaike's information criterion (AIC) is defined as

$$AIC = -2 \log(L) + \frac{2k}{T}$$

where L is the likelihood of the model, k is the total number of parameters and initial states, and T is the number of fitted values.

- ETS(A,N,N) $\alpha, \ell_0 \rightarrow k=2$
- ETS(A,Ad,N)
- AIC corrected for small sample bias (AICc) $AIC_c = AIC + \frac{2k(k+1)}{T-k-1}$
- Bayesian Information Criterion (BIC) $BIC = AIC + k[\log(T) - 2]$
- ETS () function select the model by minimizing the AICc